

Reviewed Preprint

v1 • December 8, 2025

Not revised

✉ For correspondence:

turrigiano@brandeis.edu

Competing interests: No

competing interests declared

Funding: See [page 25](#)

Reviewing editor: Brice Bathellier,

Centre National pour la Recherche

Scientifique et Technique (CNRS),

France

© 2025, Wu & Turrigiano. This article

is distributed under the terms of the

[Creative Commons Attribution](#)

[License](#), which permits unrestricted

use and redistribution provided that

the original author and source are

credited.

Learning-Associated Flexibility of Cortical Taste Coding Is Impaired in Shank3 Knockout Mice

Chi-Hong Wu, Gina G Turrigiano ✉

Department of Biology, Brandeis University, Waltham, United States

eLife Assessment

This **valuable** study provides **solid** evidence for deficits in aversive taste learning and taste coding in a mouse model of autism spectrum disorders. Specifically, the authors found that Shank3 knockout mice exhibit behavioral deficits in learning and extinction of conditioned taste aversion, and calcium imaging of the gustatory cortex identified impaired neuronal responses to taste stimuli. This paper will likely be of interest to researchers studying how learning and sensory processes are affected by genetic causes of autism spectrum disorders.

<https://doi.org/10.7554/eLife.108988.1.sa3>

Abstract

The ability to update the valence of sensory perception to influence behavior is crucial for survival. A common phenotype in autism spectrum disorders (ASDs) is defects in sensory processing, but whether these defects impair flexible sensory encoding is largely unexplored. In particular, how genetic risk factors such as *Shank3* deletion affect the adaptability of cortical taste processing and downstream behavior is unknown. To address this gap, we performed two-photon calcium imaging during a conditioned taste aversion (CTA) learning paradigm, an ethologically relevant form of associative learning that depends on taste processing in the anterior insular cortex (AIC), to examine how *Shank3* knockout alters taste-related neuronal activity in AIC and influences CTA learning. We found that AIC neurons in *Shank3* knockout mice exhibited reduced stimulus-evoked suppression and increased trial-to-trial correlated variability during the acquisition of CTA memory. These activity changes, which likely reduced signal-to-noise ratio in AIC, were associated with slower CTA acquisition in knockout mice. In both genotypes, CTA learning enhanced, while subsequent extinction reduced, taste discriminability in AIC, and both extinction and the associated reduction in discriminability were faster in knockout than in wild-type mice. Together, these results show that *Shank3* loss is associated with destabilized cortical activity dynamics in AIC, which may contribute to inefficient encoding and maintenance of learned taste aversion. These findings show that loss of *Shank3* compromises the ability of animals to update behavior to incorporate negative outcomes, and suggest this loss of flexibility may be an important feature of monogenic ASDs.

Introduction

Ethological experiences can modify fundamental features of sensory coding, such as the valence of a taste, through changes in synaptic and network functions within sensory cortices (Harris and Mrsic-Flogel, 2013 [DOI](#); Choi, 2018 [DOI](#)). This flexibility in sensory coding ensures that animals' perceptions and downstream behaviors are updated to reflect current experiences and enhance survival. Sensory processing, including taste, is often defective in patients with autism spectrum disorders (ASDs) (Leekam et al., 2007 [DOI](#); Avery et al., 2018 [DOI](#); Scheerer et al., 2021 [DOI](#); Goldschlager et al., 2025 [DOI](#)), and several rodent monogenic ASD models, including *Shank3* knockout, exhibit sensory deficits resembling those observed in patients with ASD (Kloth et al., 2015 [DOI](#); Copping et

al., 2017 [↗](#); Monteiro and Feng, 2017 [↗](#); Delling and Boeckers, 2021 [↗](#)). This altered sensory processing could contribute to downstream defects in flexible sensory coding in Shank3 knockout rodent models, but this possibility has not been directly tested. Here, we used conditioned taste aversion (CTA), an ethologically relevant form of associative learning that involves taste processing, to examine how disrupted cortical taste coding impacts the learning and extinction of aversive taste associations in Shank3 knockout mice.

CTA arises when a normally palatable taste is paired with agents (such as lithium chloride) that cause gastric malaise (Garcia et al., 1955 [↗](#)); this aversive learning is fast (requiring one or two conditioning trials) and strongly suppresses the consumption of the conditioned tastant. Like other forms of aversive learning, CTA undergoes extinction when the tastant no longer predicts the adverse outcome (Berman and Dudai, 2001 [↗](#)). Extinction occurs gradually over repeated exposures to the conditioned tastant alone; thus, during CTA and subsequent extinction, the valence of a tastant can shift from positive to negative and back to positive again in an ethologically relevant manner, providing a nice paradigm for studying the adaptability of sensory coding. CTA relies on the anterior insular cortex (AIC), where much of taste coding occurs (Katz et al., 2002 [↗](#); Fletcher et al., 2017 [↗](#); Wang et al., 2018 [↗](#); Levitan et al., 2019 [↗](#); Vincis et al., 2020 [↗](#); Chen et al., 2021 [↗](#)). CTA's acquisition and extinction induce distinct forms of plasticity within AIC (Escobar et al., 1998 [↗](#); Li et al., 2016 [↗](#); Rodríguez-Durán et al., 2017 [↗](#); Haley et al., 2020 [↗](#); Wu et al., 2021 [↗](#); Yiannakas et al., 2021 [↗](#); Kolatt Chandran et al., 2023 [↗](#)) as internal representations of taste are reshaped (Moran and Katz, 2014 [↗](#); Lavi et al., 2018 [↗](#); Arieli et al., 2022 [↗](#); Raymond et al., 2024 [↗](#)). Loss of Shank3 affects multisensory integration in the insular cortex (Gogolla et al., 2014 [↗](#)) and impairs several cellular forms of cortical synaptic and intrinsic plasticity (Bozdagi et al., 2010 [↗](#); Wang et al., 2011 [↗](#); Kouser et al., 2013 [↗](#); Raynaud et al., 2013 [↗](#); Tatavarty et al., 2020 [↗](#); Wen and Turrigiano, 2021 [↗](#)), raising the possibility that loss of Shank3 might impair CTA learning and extinction.

In this study, we employed two-photon calcium imaging and a head-restrained CTA paradigm to compare AIC activity in Shank3 knockout and wild-type mice during learning and extinction. We first discovered that Shank3 knockout mice learned CTA more slowly. During memory encoding, AIC neurons in these mutant mice were less suppressed by taste stimuli, with increased correlated variability that likely reduced the signal-to-noise ratio critical for taste processing within AIC. Although Shank3 knockout mice eventually formed CTA, their aversive memory was fragile and extinguished more rapidly than in wild-type mice. Following learning, AIC neurons in both groups became more reliable in their taste responsiveness; however, this reliability was accompanied by a gradual reduction in single-cell taste selectivity and a convergence of population activity as extinction progressed. This loss of taste selectivity and convergence of population activity was faster and more pronounced in Shank3 knockout than in wild-type mice. Finally, the ability to discriminate between tastants using population activity was enhanced by CTA and then maintained during extinction in wild-type mice but was rapidly lost during extinction in the Shank3 knockout condition. This work demonstrates that loss of Shank3 degrades the discriminability of single-cell and population activity in AIC, leading to slower learning and faster extinction of CTA. More broadly, this work shows that flexible sensory coding is defective in Shank3 knockout mice, and provides insight into how maladaptive sensory coding impacts associative learning in ASD.

Results

Shank3 knockout diminishes taste-elicited insular suppression and perturbs CTA acquisition

Loss of Shank3 is known to affect some aspects of sensory processing (Chen et al., 2020 [↗](#); Tatavarty et al., 2020 [↗](#); Balasco et al., 2022 [↗](#); Ortiz-Cruz et al., 2022 [↗](#)), but whether it disrupts ethologically relevant associative learning paradigms such as CTA that rely on sensory processing is unknown. To examine this, we performed longitudinal two-photon calcium imaging of AIC neurons from wild-type (WT) and Shank3-knockout mice (KO, Δexon 13-16, Peça et al., 2011 [↗](#)) in

a head-restrained CTA learning paradigm (Figure 1A-C). Using a gradient-index (GRIN) lens (Miura et al., 2020; Deng et al., 2021; Staszko et al., 2022) and a virally expressed calcium indicator (jRCaMP8s) driven by the pan-neuronal promoter *Synapsin* (Zhang et al., 2023), we were able to visualize calcium signals of individual neurons within deep layers of the dysgranular and agranular AIC, subregions crucial for the expression of CTA (Figure 1B, C, Yiannakas et al., 2021; Kolatt Chandran et al., 2023).

We then trained water-restricted, head-restrained mice (WT or KO littermates) to lick a motorized spout in daily sessions, each consisting of approximately 30 trials (Chen et al., 2021). This trial-based structure enabled us to control the type of the tastant animals received in each trial, as well as the duration and timing of its delivery, while simultaneously recording the corresponding neuronal calcium signals in AIC (Figure 1A, C). To ensure that animals learned CTA effectively, we subjected habituated animals to two conditioning sessions (CST1 and CST2); within each session animals received, on average, 30 deliveries of the novel tastant saccharin (conditioned stimulus, CS), followed by a LiCl injection (unconditioned stimulus, US) (referred to as the “CTA group”). A separate group of animals received saline instead of LiCl and served as the “non-learned” comparison (referred to as the “CS-only group”). In subsequent retrieval test sessions that spanned five days (T1-5), we measured the expression and extinction of CTA by giving animals saccharin (the conditioned tastant) and water (a neutral tastant) in an alternative-choice task (Chen et al., 2021; Kogan and Fontanini, 2024). While animals in the CS-only group maintained their preference for saccharin across the first and second conditioning sessions (CST1 and CST2, Figure 1D), WT mice that received LiCl significantly reduced licks to saccharin between CST1 and CST2 (Figure 1E). Although KO animals also showed a significant reduction in their saccharin consumption between CST1 and CST2 (Figure 1F), this reduction was smaller than in WT mice (Figure 1G).

To ensure reduced CTA during CST2 in KO mice was not an artifact of measuring aversion without a choice between saccharin and water, we used a modified paradigm where a separate group of animals received an alternative-choice test after CST1 (Figure 1 – figure supplement 1A). In this paradigm as well, KO mice showed a significantly higher preference for saccharin across concentrations compared to WT littermates (Figure 1 – figure supplement 1B); further, comparable lick counts in WT and KO mice suggest no major differences in the thirst state or motivation between groups (Figure 1 – figure supplement 1C, D).

Since taste responses are processed in AIC (Katz et al., 2002), we wished to know whether saccharin-elicited activity in AIC during CTA acquisition is different in WT and KO animals. We identified neurons that were activated or suppressed during various phases of CST1 and 2, and (as observed previously, Chen et al., 2021) found that AIC neurons could be influenced by acoustic cues (1-2), lick movements (2-3), and tastant exposure (3-4, Figure 1 – figure supplement 2A-C). Most of these neurons responded to more than one sensorimotor stimulus (Figure 1 – figure supplement 2D), highlighting the multi-modal and integrative nature of the insular cortex (Samuelsen et al., 2012; Gogolla et al., 2014; Vincis and Fontanini, 2016; Gogolla, 2017). In most AIC neurons, the strongest trial-averaged responses were tied to tastant delivery (Figure 1H, I, Figure 1 – figure supplement 2A, B; between yellow dashed lines marked 3 and 4); in this study, we focus on these taste-elicited responses. In both WT and KO mice that underwent CTA, we detected neuronal populations that were either suppressed (Figure 1H, I, J) or activated (Figure 1H, I, K) by tastant presentations during CST1 and CST2. There were no significant changes between CST1 and 2 for either the CTA or the CS alone groups (Figure 1 – figure supplement 2E-J). Strikingly, however, the neuronal suppression in response to saccharin was significantly reduced in the KO mice; a smaller fraction of neurons were suppressed (Figure 1L), and the average magnitude of suppression decreased (Figure 1J, M). There was a smaller reduction in the population and magnitude of activated neurons as well, but this did not reach statistical significance (Figure 1N, O). These data reveal that Shank3 KO mice have less cortical suppression during tastant delivery and exhibit reduced or slowed acquisition of CTA.

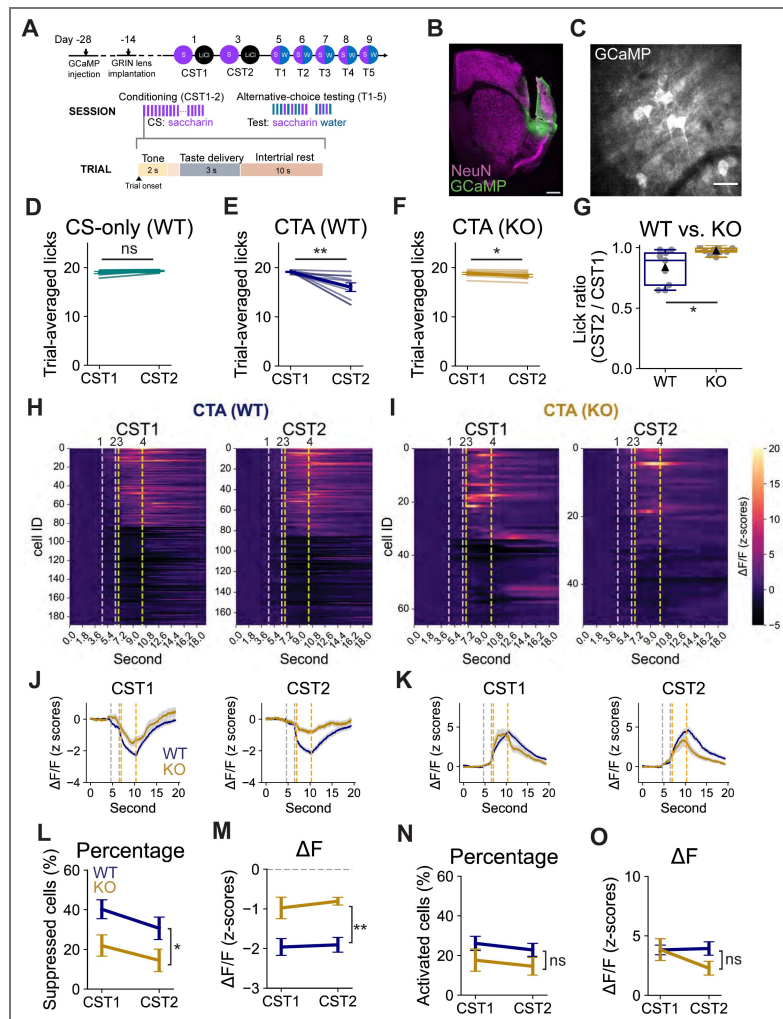


Figure 1. Shank3 knockout diminishes taste-elicited insular suppression and perturbs CTA acquisition

(A) The head-restrained aversive taste learning paradigm. CS: conditioned stimulus, 25 mM saccharin, US: unconditioned stimulus, lithium chloride (LiCl, 0.15M, 2% body weight). CST: conditioning session, T: retrieval test session. (B) Example image showing the GRIN lens placement on top of the anterior insular cortex (AIC) (green: GCaMP8s, magenta: NeuN). Scale bar: 500 μ m. (C) Example image showing the calcium signals recorded from individual AIC neurons. Scale bar: 50 μ m. (D, E, F) Trial-averaged licks during the first (CST1) and second (CST2) conditioning sessions in the CS-only (WT) group (D), the CTA (WT) group (E), and the CTA (KO) group (F) (CS-only (WT): paired t-test, $p = 0.1201$; CTA (WT): paired t-test, $p = 0.0094$; CTA (KO): paired t-test, $p = 0.0490$). Here and below where behavior data are presented: CS-only (WT), $n = 7$ animals; CTA (WT), $n = 9$ animals; CTA (KO), $n = 9$ animals. (G) The trial-averaged lick ratio of CST2 to CST1 in CTA (WT) and CTA (KO) groups. (WT vs. KO: unpaired two-tailed t-test, $p = 0.0123$). (H, I) Heatmaps showing z-scored changes in calcium responses ($\Delta F/F$) from neurons that are responsive to tastant delivery during conditioning in CTA (WT) (H, CST1: 189 cells, CST2: 172 cells) and CTA (KO) (I, CST1: 66 cells, CST2: 52 cells) groups. 1: cue onset; 2: spout in place; 3: tastant delivery starts; 4: tastant delivery ends. The 3-second taste delivery is marked by yellow dashed lines. In each heatmap, cells are sorted in descending temporal order based on whether their activity is activated or suppressed. Here and below where calcium imaging data are presented: CS-only (WT), $n = 7$ animals; CTA (WT), $n = 8$ animals; CTA (KO), $n = 7$ animals. (J) Calcium traces averaged across suppressed neurons in the CTA (WT) and CTA (KO) groups during CST1 (left) and CST2 (right). (K) Calcium traces averaged across activated neurons in the CTA (WT) and CTA (KO) groups during CST1 (left) and CST2 (right). (L) Change in the percentage of suppressed neurons from CST1 to CST2 (WT vs. KO, repeated-measures mixed ANOVA, genotype: $p = 0.0218$, session: $p = 0.0393$, interaction: 0.7819). (M) Change in the calcium fluorescence ($\Delta F/F$) of suppressed neurons from CST1 to CST2. Dashed lines mark the value 0 (WT vs. KO, repeated-measures mixed ANOVA, genotype: $p = 0.0011$, session: $p = 0.7356$, interaction: 0.8672). (N) Change in the percentage of activated neurons from CST1 to CST2 (WT vs. KO, repeated-measures mixed ANOVA, genotype: $p = 0.1391$, session: $p = 0.2590$, interaction: 0.9705). (O) Change in the calcium fluorescence ($\Delta F/F$) of activated neurons from CST1 to CST2 (WT vs. KO, repeated-measures mixed ANOVA, genotype: $p = 0.3404$, session: $p = 0.2664$, interaction: 0.1295). Error bars in this figure and below indicate s.e. (standard error of the mean).

Shank3 knockout increases the correlated variability in AIC during CTA acquisition

We wondered what other features of AIC activity during CTA acquisition might differ between WT and Shank3 KO mice. The correlated variability between neurons – the trial-to-trial co-fluctuation of their activity over repeated presentations of the same stimulus – has been shown to limit the information-encoding capacity of neuronal networks (Moreno-Bote et al., 2014 [DOI](#); Bartolo et al., 2020 [DOI](#); Rumyantsev et al., 2020 [DOI](#); Kafashan et al., 2021 [DOI](#); Hazon et al., 2022 [DOI](#)). We thus wondered whether Shank3 KO might impede CTA by increasing the correlated variability in AIC during tastant presentations. We first characterized the trial-to-trial coactivity between AIC neurons by measuring the pairwise correlations of their mean calcium responses across CS (saccharin) presentations (see methods). There was a rightward shift in the cumulative distribution function of these pairwise correlations in KO compared to WT animals, which was apparent in both conditioning sessions (Figure 2A, B [DOI](#)). Comparing the mean pairwise correlation calculated by animal confirmed a persistent increase in the average strength of coactivity in the KO mice across conditioning sessions (Figure 2C [DOI](#)).

Neuronal coactivity could result from “signal” correlations, which reflect the similarity of average stimulus-tuning properties between neurons, and “noise” correlations that measure trial-to-trial activity co-fluctuations between neurons independent of stimulus tuning (Panzeri et al., 2022 [DOI](#)). To determine if the “noise” component is the driving force of the increased coactivity observed in the KO mice, we performed a different correlation analysis that allowed us to separate signal-induced correlations from the residual (noise) correlations (Marks and Goard, 2021 [DOI](#)). For a given neuron, we first obtained the stimulus-induced signal by averaging response traces (3 seconds) across trials (Figure 2D [DOI](#), top). We then estimated the neuron’s trial-to-trial variability (‘noise’) by subtracting this averaged ‘signal’ from individual trials and concatenating the residuals (Figure 2D [DOI](#), bottom). The signal correlation was then quantified by calculating the pairwise correlations of the averaged signal, while the noise correlation was calculated as the pairwise correlations in the trial-to-trial variability. While the signal correlation was reduced from CST1 to CST2, there was no difference between WT and KO mice (Figure 2E, F [DOI](#)). By contrast, the correlated variability was significantly higher in the KO mice (Figure 2G, H [DOI](#)). The activity correlations between neurons generally decrease when the intraneuronal distances grow wider (Rothschild et al., 2010 [DOI](#); Rosenbaum et al., 2017 [DOI](#); Khoury et al., 2022 [DOI](#)). To determine the spatial extent of enhanced correlations in Shank3 KO mice, we measured correlated variability as a function of the inter-somatic distance between neuron pairs (Figure 2I [DOI](#)); strikingly, the increased coactivity was observed even in neuronal pairs > 300 μm apart (Figure 2I [DOI](#)), indicating a network-wide, rather than local, origin. Thus, in the Shank3 KO mice, there is a widespread increase in correlated variability between AIC neurons, suggesting there is more “noise” in insular cortical circuits.

Finally, to determine if this coactivity is stimulus-dependent, we repeated the correlation analysis using the mean calcium responses two seconds prior to cue onset (Lee et al., 2020 [DOI](#)). Measuring the trial-to-trial correlations of this spontaneous activity showed that coactivity was enhanced in KO animals even at baseline (Figure 2 – figure supplement 1A-C [DOI](#)). Furthermore, as for taste-elicited activity, this increased baseline coactivity had a broad spatial extent (Figure 2 – figure supplement 1D [DOI](#)). Altogether, these analyses demonstrate that Shank3 KO mice have increased network-wide, stimulus-independent correlated variability within AIC. This heightened background noise is likely to reduce the signal-to-noise ratio during tastant processing, crucial for the induction of CTA.

Extinction of CTA in Shank3 knockout mice is faster than in wild-type littermates

The reduced expression of CTA in KO mice after the first conditioning session (Figure 1 [DOI](#)) could be due to a reduction or a delay in the learning process. To differentiate between these possibilities, we measured taste aversion after the second round of conditioning (after CST2), by

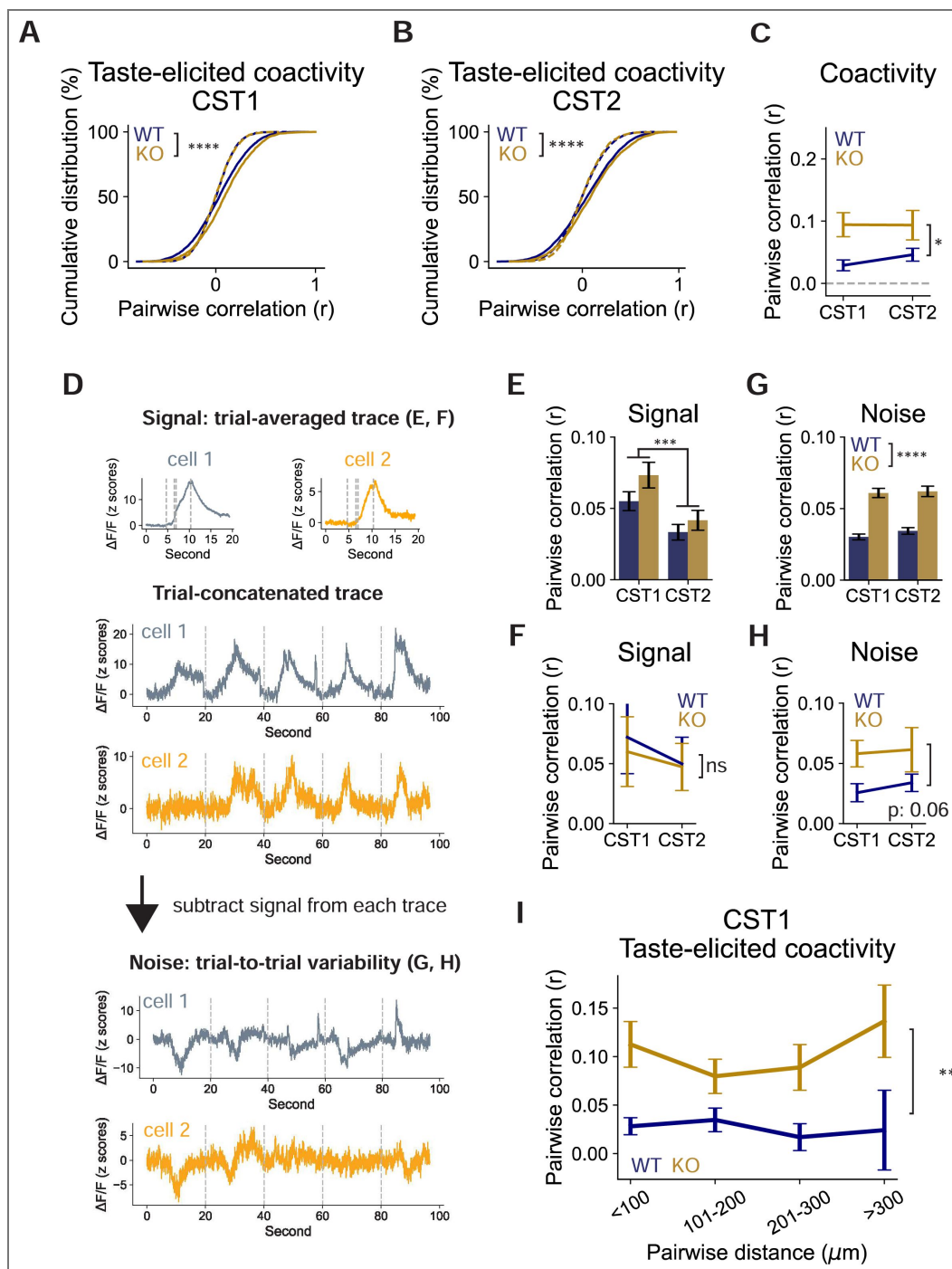


Figure 2. Shank3 knockout increases the correlated variability in AIC during CTA acquisition

(A, B) Cumulative distribution of taste-elicited coactivity (Pearson coefficients) between neurons during CST1 (A) and CST2 (B). Dashed lines indicate the shuffled data (CST1: WT, $n = 6201$ pairs, KO, $n = 2645$ pairs, WT vs. KO, Kolmogorov-Smirnov test, $p < 0.0001$; CST2: WT, $n = 8476$ pairs, KO, $n = 2847$ pairs, WT vs. KO, Kolmogorov-Smirnov test, $p < 0.0001$). (C) Change in the average taste-elicited coactivity from CST1 to CST2, plotted by animal. The dashed line indicates the chance-level correlation. (WT vs. KO, repeated-measures mixed ANOVA, genotype: $p = 0.0142$, session: $p = 0.4198$, interaction: 0.4146). (D) Schematic showing how the signal and noise correlations are computed. (E, F) Change in the signal correlation from CST1 to CST2, plotted by neuronal pair (E) or by animal (F) (E: WT: CST1, $n = 6201$ pairs, CST2, $n = 8476$ pairs; KO: CST1, $n = 2645$ pairs, CST2, $n = 2847$ pairs, two-way ANOVA, genotype: $p = 0.1134$, session: $p = 0.0009$, interaction: 0.5469 ; F: WT vs. KO, repeated-measures mixed ANOVA, genotype: $p = 0.8143$, session: $p = 0.4007$, interaction: 0.8213). (G, H) Change in the correlated variability from CST1 to CST2, plotted by neuronal pair (G) or by animal (H) (G: two-way ANOVA, genotype: $p < 0.0001$, session: $p = 0.2274$, interaction: 0.6112 ; H: WT vs. KO, repeated-measures mixed ANOVA, genotype: $p = 0.0617$, session: $p = 0.4017$, interaction: 0.7188). (I) Average taste-elicited coactivity in CST1, grouped by inter-neuronal distance and plotted by animal (WT vs. KO, repeated-measures mixed ANOVA, genotype, $p = 0.0050$, distance, $p = 0.7643$, interaction: $p = 0.5895$).

comparing the ratio of licks to saccharin and water during the first retrieval test session (T1, Figure 1A). We found that both WT and KO mice exhibited strong and comparable CTA during T1 (Figure 3A-D), indicating that CTA formation is slower but still significant in KO mice.

As expected, WT animals underwent gradual memory extinction between T1 and T5 (Berman and Dudai, 2001; Yiannakas and Rosenblum, 2017), with their preference for saccharin slowly returning close to baseline by T4-5 (Figure 3B, D). In contrast, KO mice underwent much faster extinction, with their saccharin consumption returning to baseline by T2 (Figure 3C, D). To calculate the rate of extinction, we fitted a logistic curve to the lick-ratio data for each animal and estimated the rate of memory extinction by taking the derivative of the fit (Figure 3E, F, Kogan and Fontanini, 2024). This analysis revealed distinct extinction dynamics between WT and KO mice (Figure 3E, F): as expected, the lick ratio of saccharin to water increased gradually over time in WT (Figure 3E, top) while rising steeply in KO mice, with a much higher peak rate (Figure 3F, G). Thus, Shank3 KO mice not only form CTA more slowly but also undergo faster CTA extinction than their WT littermates.

CTA learning increases the response reliability of individual AIC neurons

The stimulus responsiveness of individual sensory neurons is prone to drift over time (Deitch et al., 2021; Marks and Goard, 2021; Schoonover et al., 2021). We wondered whether CTA might stabilize taste-elicited insular responses, as has been suggested in other sensory modalities (Poort et al., 2015; Glas et al., 2021; Taylor et al., 2021; Biane et al., 2023; Pancholi et al., 2023). To test this, we identified AIC neurons that could be recorded in each retrieval test session (Figure 4A, see methods) and tracked their daily responsiveness to taste. We first focused on cells that were activated by saccharin in at least one session and found that in the CS-only group, a majority of neurons dropped in and out of the population responses (Figure 4B). In contrast, both WT and KO CTA groups had a higher proportion of neurons reliably activated by saccharin across days (Figure 4C, D). While no neurons in the CS-only group were active across all five sessions, 16 (WT) and 15 (KO) percent of neurons in the CTA conditions were responsive throughout (Figure 4E). The same pattern was found for AIC neurons that were responsive to water, an independent and neutral taste modality that induces distinct neuronal activities in AIC (Figure 4F-I, Katz et al., 2001; Rosen et al., 2010). Hence, associative learning enhances the reliability of AIC neurons' responsiveness in both genotypes.

The stimulus selectivity of individual AIC neurons degrades faster in Shank3 knockout mice during memory extinction

The feature selectivity of individual neurons within primary sensory cortices is dynamic and experience-dependent (Chéreau et al., 2020). We reasoned that taste selectivity – the “breadth of tuning” (Levitani et al., 2019) – of individual AIC neurons might evolve during CTA extinction, and that this process might be accelerated in Shank3 KO mice to promote faster extinction. To test this, we identified neurons activated by water or saccharin delivery, and in test sessions estimated the percentage of neurons that responded to one or both stimuli (Figure 5A). Our analysis revealed a comparable proportion of nonselective AIC neurons (~40 percent) in both genotypes during the first test session (T1, Figure 5B). During extinction, the percentage of nonselective neurons gradually increased in the WT group and reached ~50 percent by T5 (Figure 5B); this increase was faster and stronger in KO mice, reaching ~65 percent by T5 (Figure 5B).

While binarizing calcium signals into “active” and “non-active” is widely used to describe single-cell stimulus-elicited responses (Zhang and Li, 2018; Gehrlach et al., 2019; Wang et al., 2020; Chen et al., 2021; Staszko et al., 2022; Biane et al., 2023), it ignores changes in response amplitude that could encode stimulus-related information. We therefore performed a complementary analysis and calculated a discriminability index based on the amplitude differences between each neuron's trial-averaged calcium responses to saccharin and water (see methods, Banerjee et al., 2020; Lee et al., 2021; Kong et al., 2023). A more positive index

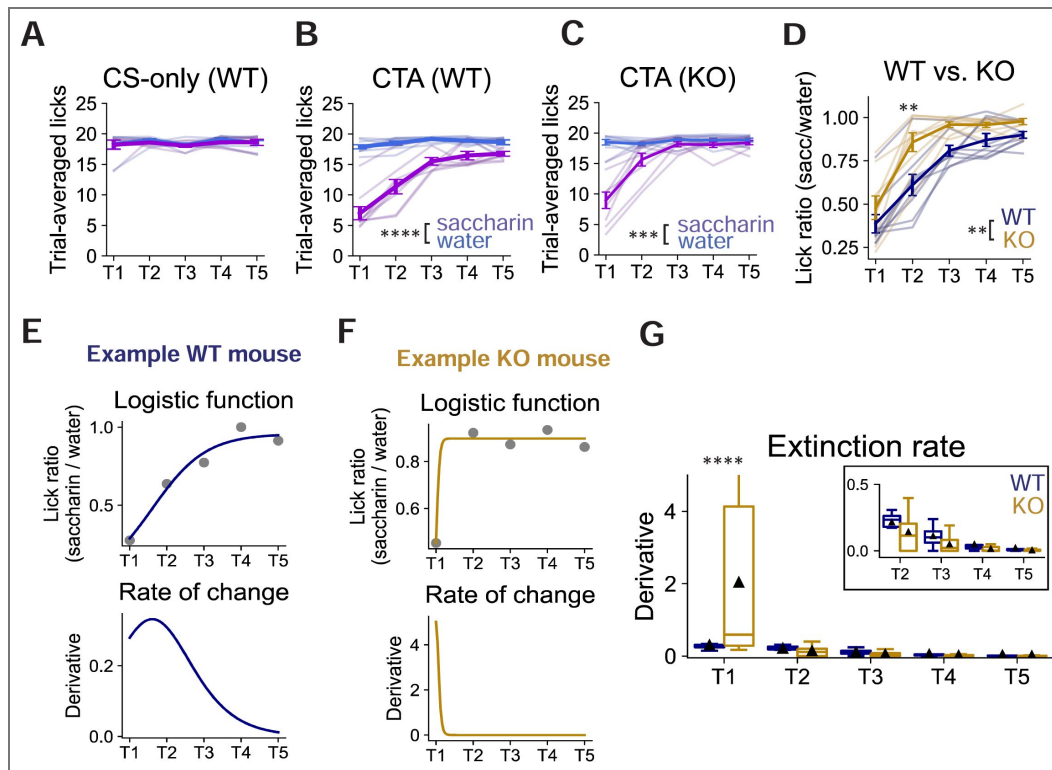


Figure 3. Extinction of CTA in Shank3 knockout mice is faster than in wild-type littermates

(A, B, C) Trial-averaged licks of saccharin (violet) and water (blue) in the CS-only (WT) (A), CTA (WT) (B), and CTA (KO) (C) groups throughout the test sessions (CS-only (WT): repeated-measures ANOVA, taste, $p = 0.2530$, session, $p = 0.4158$, interaction, $p = 0.4515$; CTA (WT): repeated-measures ANOVA, taste, $p < 0.0001$, session, $p < 0.0001$, interaction, $p < 0.0001$; CTA (KO): repeated-measures ANOVA, taste, $p = 0.0002$, session, $p < 0.0001$, interaction, $p < 0.0001$). (D) Trial-averaged lick ratios of saccharin to water throughout the test sessions in CTA (WT) and CTA (KO) groups (WT vs. KO: repeated-measures mixed ANOVA, genotype, $p = 0.0069$, session, $p < 0.0001$, interaction, $p = 0.0691$, post-hoc Tukey test, WT vs. KO, T1 $p = 0.8669$, T2 $p = 0.0032$, T3 $p = 0.2705$, T4 $p = 0.9014$, T5 $p = 0.9531$). (E, F) A fitted logistic curve (top) and its derivative (bottom) describing the relationship between lick ratios and test sessions in an example WT mouse (E) and an example KO mouse (F). (G) Average rate of change (derivative) at each test session. The triangle indicates the data mean. (WT vs. KO: repeated-measures mixed ANOVA, genotype, $p = 0.0271$, session, $p < 0.0001$, interaction, $p = 0.0003$; post-hoc Tukey test, WT vs. KO, T1 $p < 0.0001$, T2 $p = 1.0000$, T3 $p = 1.0000$, T4 $p = 1.0000$, T5 $p = 1.0000$).

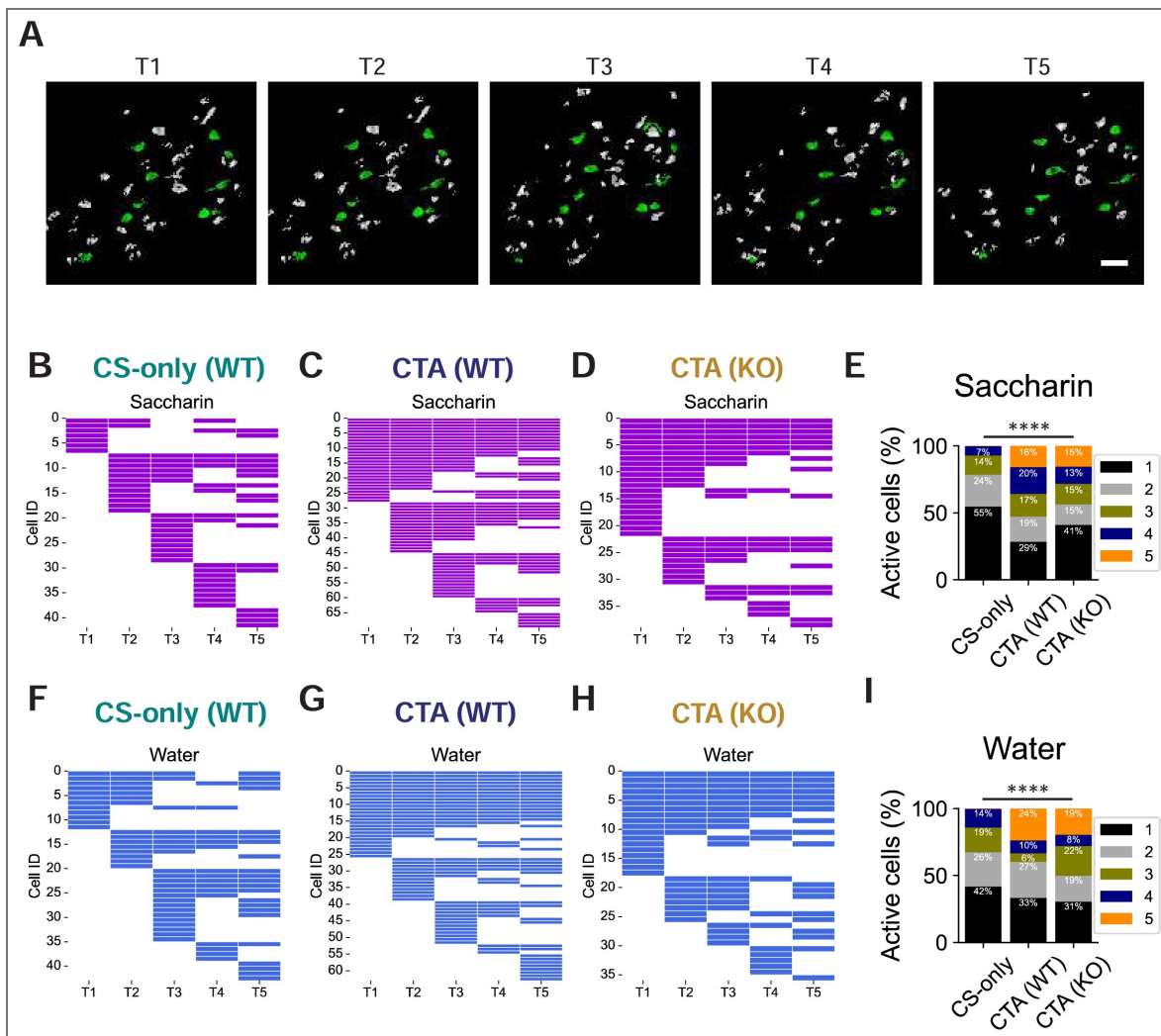


Figure 4. CTA learning increases the response reliability of individual AIC neurons

(A) Example images showing neurons present in all five retrieval test sessions (labeled green). Scale bar: 20 μ m. **(B, C, D)** Lifetime plot showing significant responses of the same neurons across test sessions (T1-5) in the CS-only (WT) group (B, n = 42 cells), the CTA (WT) group (C, n = 70 cells), and the CTA (KO) group (D, n = 39 cells). Violet labels activated states, and white labels nonactivated states. **(E)** Percentage of neurons active for 1, 2, 3, 4, and 5 sessions in (B-D) (Chi-square test, $p < 0.0001$). **(F, G, H)** Lifetime plot showing the water-responsiveness of the same neurons across test sessions (T1-5) in the CS-only (WT) group (F, n = 43 cells), the CTA (WT) group (G, n = 63 cells), and the CTA (KO) group (H, n = 36 cells). Blue labels activated states, and white labels nonactivated states. **(I)** Percentage of neurons active for 1, 2, 3, 4, and 5 sessions in (F-H) (Chi-square test: $p < 0.0001$).

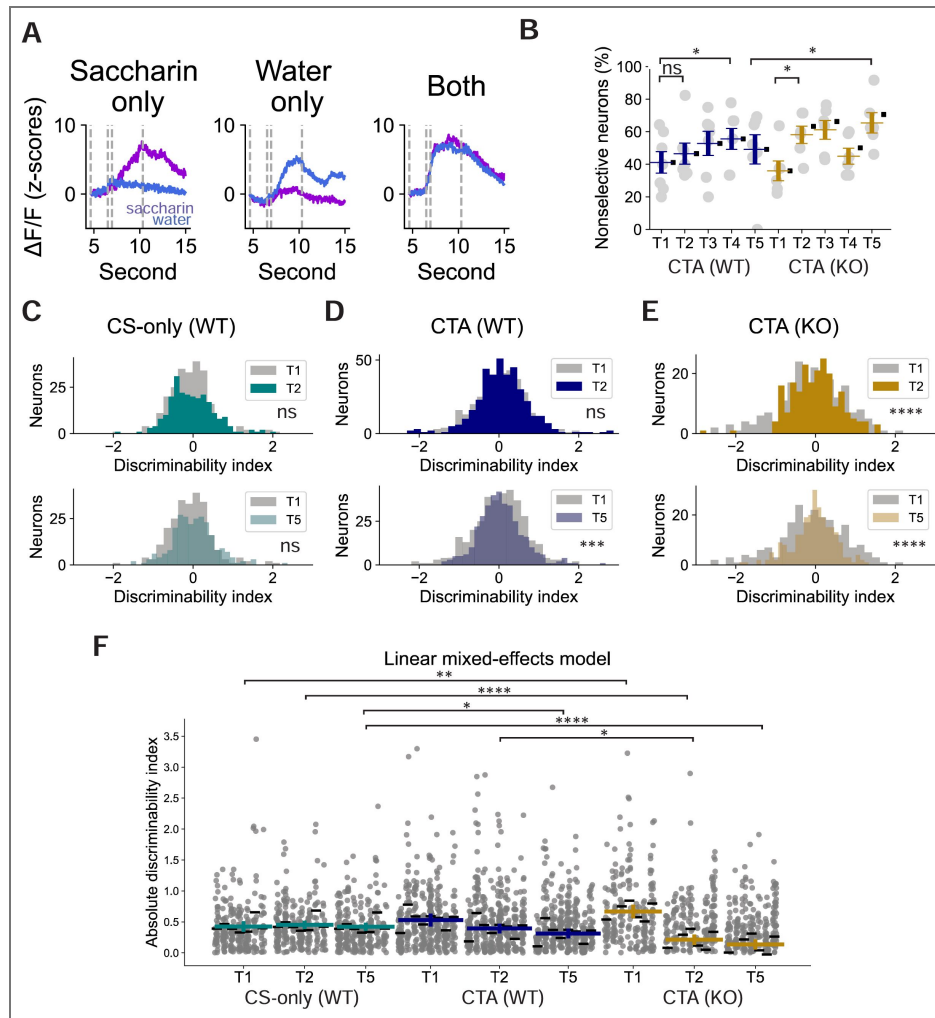


Figure 5. The stimulus selectivity of individual AIC neurons degrades faster in Shank3 knockout mice during CTA extinction

(A) Calcium traces of example neurons activated by saccharin only (left), water only (middle), or both (right). Violet marks the calcium responses to saccharin, and blue marks the responses to water. **(B)** The percentages of nonselective neurons across test sessions and genotypes. Each data point represents one animal. The colored bar marks the data-averaged value and its standard error. The square indicates the fitted mean in each condition based on the linear mixed-effects model, with session and genotype set as fixed parameters and individual animal as random effect (WT vs. KO, linear mixed-effects model, genotype (in reference to WT): KO, $\beta = -5.146$, $p = 0.554$, session (in reference to T1): T2, $\beta = 5.410$, $p = 0.438$, T3, $\beta = 11.692$, $p = 0.093$, T4, $\beta = 14.433$, $p = 0.038$, T5, $\beta = 8.045$, $p = 0.248$, interaction: KO $\times$ T2, $\beta = 16.729$, $p = 0.103$, KO $\times$ T3, $\beta = 13.466$, $p = 0.189$, KO $\times$ T4, $\beta = -5.420$, $p = 0.597$, KO $\times$ T5, $\beta = 21.409$, $p = 0.037$, random effect, variance, 74.187; post hoc t-test, KO, T1 vs. T2, $p = 0.0218$). **(C, D, E)** Histograms comparing the distributions of the discriminability index between T1 and T2 (top) and between T1 and T5 (bottom) in the CS-only (WT) group (C, T1: 243 cells, T2: 238 cells, T5: 250 cells, F-test with Bonferroni correction, T1 vs. T2 $p = 0.5979$, T1 vs. T5 $p = 0.2110$), the CTA (WT) group (D, T1: 355 cells, T2: 382 cells, T5: 358 cells, F-test with Bonferroni correction, T1 vs. T2 $p = 0.6687$, T1 vs. T5 $p = 0.0001$), and the CTA (KO) group (E, T1: 212 cells, T2: 212 cells, T5: 241 cells, F-test with Bonferroni correction, T1 vs. T2 $p < 0.0001$, T1 vs. T5 $p < 0.0001$). **(F)** Comparison of absolute discriminability index across CS-only (WT), CTA (WT), and CTA (KO) groups. Each data point indicates a single-cell value, grouped by animal. Black bars indicate the “predicted” subject-level mean value from the mixed-effects model accounting for both fixed (session, genotype) and random (animal) effects. The colored bars mark the “predicted” marginal means from fixed effects (session and genotype) (linear mixed-effects model, group (in reference to CS-only, T1): CTA (WT), $\beta = 0.1058$, $p = 0.1538$, CTA (KO), $\beta = 0.2442$, $p = 0.0018$; session (in reference to CS-only, T1): T2, $\beta = 0.0271$, $p = 0.4598$, T5, $\beta = -0.0035$, $p = 0.9239$; interaction: CTA (WT) $\times$ T2, $\beta = -0.0577$, $p = 0.2203$, CTA (WT) $\times$ T5, $\beta = -0.1082$, $p = 0.0213$, CTA (KO) $\times$ T2, $\beta = -0.2393$, $p < 0.0001$, CTA (KO) $\times$ T5, $\beta = -0.2856$, $p < 0.0001$). Planned post hoc comparisons are tested using linear contrasts on the fitted model with Benjamini-Hochberg correction for multiple comparisons (T2 vs. T1: CTA (WT), $p = 0.6455$, CTA (KO), $p = 0.9742$; WT vs. KO: T1, $p = 0.6455$, T2, $p = 0.0301$, T3, $p = 0.1040$).

indicates the neuron has a higher response preference for saccharin, while neurons with a more negative index respond more strongly to water. We first quantified the spread of the distribution by pooling the discriminability indices from all recorded neurons, where a narrowing of the distribution indicates that neurons within the population become more similar in their response preference. In the CS-only group, the variance was similar for the first and last test sessions (Figure 5C). In WT mice that underwent CTA, there was a gradual reduction in the variance (narrowing) of the distribution from T1 to T5 (Figure 5D), and this decrease was already apparent during T2 in KO mice (Figure 5E). Calculating the absolute discriminability index (which reflects the degree of a neuron's selectivity regardless of the taste stimulus it responds to) revealed that by T5, the mean absolute index was significantly smaller in the CTA group compared to the CS-only group (CS-only (WT) vs. CTA (WT) at T5, $p = 0.0213$, Figure 5F). This reduction in taste discriminability was again faster and more pronounced in KO mice (CTA (WT) vs. CTA (KO) at T2: $p = 0.0301$, Figure 5F). This reduction in discriminability was not due to changes in the number and response amplitude of both taste-activated and suppressed AIC neurons, as these remained stable throughout extinction (Figure 5 – figure supplement 1). Thus, although the responses of individual AIC neurons become more reliable after CTA, they gradually provide less information about taste identity during CTA extinction, and this process is accelerated in the KO condition.

Learning improves population coding of taste in AIC, but this enhancement rapidly decays in Shank3 knockout mice

To determine whether the experience-dependent changes in taste discriminability are reflected in population-level activity, we calculated the Euclidean distance between population vectors in response to water or saccharin exposures; this is a widely used measure of the similarity between population neuronal responses to a pair of stimuli (Levitan et al., 2019; Arieli et al., 2022). In the CS-only group, the population responses to water and saccharin remained highly similar, as shown by the consistently small distance between the two (Figure 6A). By comparison, CTA roughly doubled the distance (dissimilarity) between the two taste-elicited population responses, and while this distance decreased during extinction, it remained higher than for the CS-only group (Figure 6A). To determine whether these population differences are sufficient to reliably classify tastants, we trained a support vector classifier to predict the taste class (water or saccharin) based on the ΔF values in the neuronal feature space (see methods, Bishop, 2006; Woods et al., 2020; Raymond et al., 2024). We trained and evaluated a classifier for each animal within each session to capture potential behavioral variability (Twarkowski et al., 2022). While decoding accuracy was close to chance in the CS-only group, CTA learning led to significantly better classification performances across sessions (Figure 6B–C). The distance between population responses to water and saccharin was reduced faster and to a larger degree in KO mice (Figure 6D, linear mixed-effects model: T1 vs. T2 in WT, $\beta = -0.099$, $p = 0.697$; T2 WT vs. KO, $\beta = -0.901$, $p = 0.015$; post hoc paired t-test: T1 vs. T2 in KO, $p = 0.009$), and this could also be seen in degraded decoding accuracy in these mutant mice (Figure 6E, F). Thus, population-level, discriminative taste representations emerge in AIC following associative learning, but Shank3 KO enhances the decay of this representation during extinction.

Lastly, we examined the stability of taste representations in AIC during memory extinction (across sessions) by using the classifiers trained on T1 responses to predict taste classes in the subsequent sessions (T2–5). In the CS-only group, the classifier performed poorly across sessions (Figure 6G). In contrast, in the CTA (WT) condition, the cross-session classifier performed well during the early stages of memory extinction, and the accuracy gradually decreased (T1 vs. T5, $p = 0.037$, Figure 6G, H). Finally, consistent with our previous findings, in the CTA (KO) condition, the decoding accuracy declined sharply from T1 to T2 (Figure 6G, H). In summary, taste representations in AIC become and remain discriminative after learning. In WT animals, these representations continue to diverge from those during the early memory retrieval as CTA is extinguished, and in KO animals, this shift is faster.

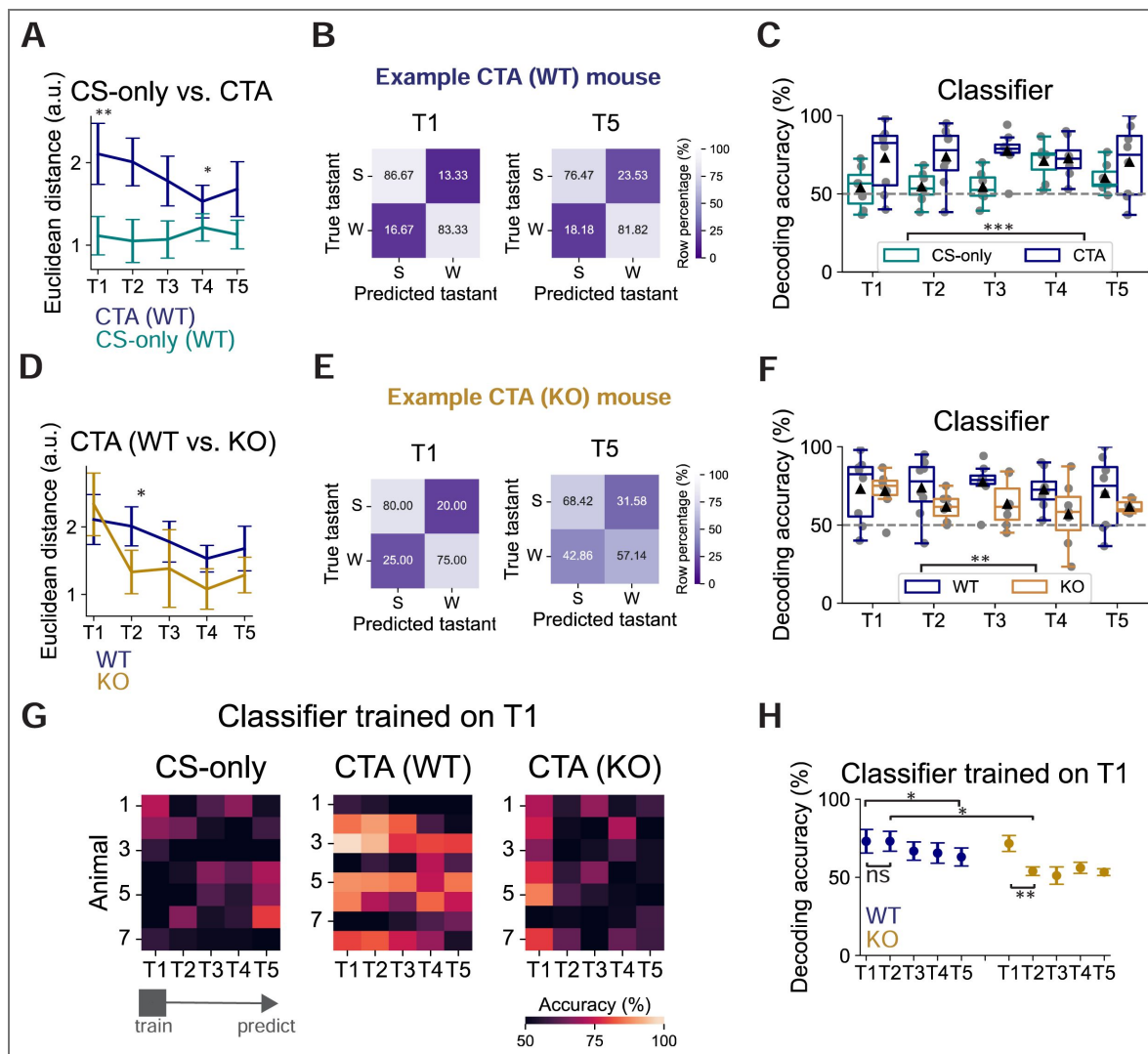


Figure 6. Learning improves population coding of taste in AIC, but this enhancement rapidly decays in Shank3 knockout mice

(A) Comparison of the Euclidean distance between population responses to water and saccharin between CTA (WT) and CS-only (WT) groups throughout test sessions (linear mixed-effects model, group (in reference to CS-only): CTA, $\beta = 0.992$, $p = 0.005$, session (in reference to T1): T2, $\beta = -0.065$, $p = 0.760$, T3, $\beta = -0.046$, $p = 0.829$, T4, $\beta = 0.101$, $p = 0.636$, T5, $\beta = 0.013$, $p = 0.950$, interaction: CTA $\times$ T2, $\beta = -0.033$, $p = 0.910$, CTA $\times$ T3, $\beta = -0.282$, $p = 0.336$, CTA $\times$ T4, $\beta = -0.680$, $p = 0.020$, CTA $\times$ T5, $\beta = -0.441$, $p = 0.132$, random effect, variance, 0.312). (B) Diagram showing that training and prediction are performed within each test session. (C) Comparison of decoding accuracies of the linear classifiers trained and tested in each session for the CS-only (WT) and the CTA (WT) groups. The dashed line indicates the chance-level predictions (CS-only (WT) vs. CTA (WT): two-way ANOVA, group, $p = 0.0001$, session, $p = 0.6637$, interaction, $p = 0.3273$). (D, E) Confusion matrices showing the prediction performances in T1 (left) and T5 (right) in an example CTA (WT) mouse (D) and an example CS-only (WT) mouse (E). (F) Comparison of the Euclidean distance between population responses to water and saccharin between CTA (WT) and CTA (KO) groups throughout test sessions (linear mixed-effects model, genotype (in reference to WT): KO $\beta = 0.225$, $p = 0.625$, session (in reference to T1): T2, $\beta = -0.099$, $p = 0.697$, T3, $\beta = -0.328$, $p = 0.195$, T4, $\beta = -0.578$, $p = 0.022$, T5, $\beta = -0.427$, $p = 0.091$, interaction: KO $\times$ T2, $\beta = -0.901$, $p = 0.015$, KO $\times$ T3, $\beta = -0.621$, $p = 0.094$, KO $\times$ T4, $\beta = -0.674$, $p = 0.069$, KO $\times$ T5, $\beta = -0.618$, $p = 0.095$, random effect, variance, 0.538; post hoc t-test, KO, T1 vs. T2, $p = 0.0089$). (G) Comparison of decoding accuracies of the linear classifiers trained and tested in each session for the CTA (WT) and the CTA (KO) groups. (WT vs. KO: two-way ANOVA, genotype, $p = 0.0078$, session, $p = 0.7293$, interaction, 0.7757). (H) Confusion matrices showing the prediction performances in T1 (left) and T5 (right) in an example CTA (KO) mouse. (I) Heatmaps of decoding accuracies using the classifiers trained on T1 responses and tested across sessions. Each row indicates an animal. (J) Quantification of the decoding accuracies in (I), comparing CTA (WT) and CTA (KO) groups (linear mixed-effects model, genotype (in reference to WT): KO, $\beta = -1.417$, $p = 0.846$, session (in reference to T1): T2, $\beta = 0.031$, $p = 0.995$, T3, $\beta = -6.286$, $p = 0.214$, T4, $\beta = -7.545$, $p = 0.136$, T5, $\beta = -10.031$, $p = 0.047$, interaction: KO $\times$ T2, $\beta = -17.732$, $p = 0.017$, KO $\times$ T3, $\beta = -14.260$, $p = 0.054$, KO $\times$ T4, $\beta = -8.013$, $p = 0.279$, KO $\times$ T5, $\beta = -8.261$, $p = 0.264$, random effect, variance, 96.479; post hoc t-test, KO, T1 vs. T2, $p = 0.0063$).

Discussion

Flexible sensory coding ensures that perceptions are updated to reflect their value to an organism, to enable adaptive behaviors. While sensory coding deficits have been reported in mouse models of ASD, including Shank3 KO (Monday et al., 2023 [↗](#)), it is unknown whether flexible coding is impaired in ethological learning paradigms, such as CTA. Here, we used 2-photon calcium imaging and a head-restrained CTA learning paradigm to address this question in Shank3 WT and KO mice. We show that neuronal ensembles in AIC, the cortical region critical for taste processing, have higher correlated noise and lower taste discriminability in Shank3 KO compared to WT littermates, and these changes are correlated with slower CTA learning, indicating that Shank3 KO animals require repeated exposure to aversive experiences to enable updating of taste valence. Once CTA is established, responses of AIC neurons in both wild-type and Shank3 KO mice are initially able to encode and discriminate between different tastants, both at the single-cell and population levels. However, during repeated exposure to the conditioned tastant, taste discriminability in Shank3 KO mice degrades faster than in their wild-type counterparts, and this coincides with the premature extinction of CTA. This work demonstrates that loss of Shank3 compromises the flexibility of sensory coding, and thus the ability of these animals to adapt their behavior in response to negative outcomes.

The increased correlated variability between AIC neurons observed in Shank3 KO mice during CTA learning may impede information encoding (Moreno-Bote et al., 2014 [↗](#); Bartolo et al., 2020 [↗](#); Romyantsev et al., 2020 [↗](#); Kafashan et al., 2021 [↗](#); Hazon et al., 2022 [↗](#)) and contribute to the slower acquisition of CTA. Reduced cortical feedback inhibition has been shown to increase the correlated variability (noise correlation) in the cortex (Stringer et al., 2016 [↗](#)). Indeed, we detected reduced neuronal suppression in the AIC of Shank3 KO mice, and because Shank3 deletion impairs cortical interneuron functions (Lee et al., 2017 [↗](#); Chen et al., 2020 [↗](#); Pagano et al., 2023 [↗](#)), the observed heightening of correlated variability could stem from weakened local inhibition within the AIC network. Heightened noise correlation has been observed across sensory cortices in different animal models of ASD, including the somatosensory and visual cortices of mice with cell-type-specific deletion of Syngap1 and MeCP2, respectively (Banerjee et al., 2016 [↗](#); Zhao and Kwon, 2023 [↗](#)), but the spatial extent of the noise correlations in these mutant animals has not been examined. Because the increased correlated variability in Shank3 KO mice occurs over a large spatial scale, it could reflect stimulus-independent, correlated activity of divergent inputs to AIC. Regardless of the source, this increased correlated noise is likely to degrade taste-elicited responses and thus hinder CTA encoding. Our behavioral results are also consistent with a recent study showing a slower acquisition of CTA in a pharmacologically induced rat model of autism (Pal et al., 2025 [↗](#)). Taken together with our finding that excessive cortical noise correlations impede discriminative sensory learning in Shank3 KO mice, these observations suggest that a reduced signal-to-noise ratio driven by high correlated variability in the sensory cortex is a common mechanism underlying maladaptive sensory processing in ASD.

AIC neurons exhibit a wide range of tuning properties (Fletcher et al., 2017 [↗](#); Levitan et al., 2019 [↗](#); Chen et al., 2021 [↗](#); Staszko et al., 2022 [↗](#)), and we found that, similar to neurons in the primary visual and somatosensory cortices (Poort et al., 2015 [↗](#); Khan et al., 2018 [↗](#); Chéreau et al., 2020 [↗](#)), the stimulus selectivity (response preference) in AIC can be dynamically adjusted at the cellular level, from being selective to nonselective during CTA extinction. While CTA learning can still initiate selective response preferences in AIC neurons of Shank3 KO mice, this stimulus selectivity was rapidly degraded during the premature extinction of CTA memory. The maintenance and extinction of a faithful CTA memory rely on both Hebbian and homeostatic synaptic plasticity mechanisms within AIC (Escobar et al., 1998 [↗](#); Li et al., 2016 [↗](#); Rodríguez-Durán et al., 2017 [↗](#); Haley et al., 2020 [↗](#); Wu et al., 2021 [↗](#)). Because Shank3 KO is known to perturb several of these critical plasticity processes (Bozdagi et al., 2010 [↗](#); Raynaud et al., 2013 [↗](#); Tatavirt et al., 2020 [↗](#)), rapidly reduced stimulus selectivity of AIC neurons could result from defects in one or more of these local plasticity mechanisms. An alternative, though not mutually exclusive, possibility is that the rapid loss of stimulus selectivity and CTA memory results from the impairment in other brain regions that project to AIC, such as BLA and the ventral tegmental area

(VTA) (Levitán et al., 2020 [↗](#); Gil-Lievana et al., 2022 [↗](#)). Hebbian plasticity can be induced at synaptic inputs from BLA to AIC during CTA learning (Escobar et al., 1998 [↗](#); Rodríguez-Durán et al., 2017 [↗](#); Haley et al., 2020 [↗](#)), and dopaminergic projections from VTA facilitate the induction of NMDAR-dependent LTP within AIC (Rodríguez-Durán et al., 2023 [↗](#)). Thus, Shank3 KO could disrupt function in these upstream brain regions to impair plastic changes in AIC during extinction. This possibility is consistent with the finding that Shank3 down-regulation in VTA accelerates extinction of socially induced conditioned place preference (CPP) memory (Bariselli et al., 2018 [↗](#)).

During CTA extinction, population responses to the neutral and conditioned taste stimuli in AIC gradually become more similar. Because CTA learning can be suppressed by the familiarity of a tastant (De la Casa and Lubow, 1995 [↗](#)), we were not able to compare the population responses to different taste stimuli before and after CTA learning; however, even after extinction these population responses were more distinct in the CTA than the CS-only group (demonstrated by the Euclidean distance and the classifier prediction), suggesting that taste coding in AIC does not revert to its naïve state even when the behavioral aversion is gone. This is consistent with previous *in vivo* electrophysiology demonstrating that, during CTA extinction, the population-level taste coding in AIC continuously evolves but does not return to the pre-CTA state (Moran and Katz, 2014 [↗](#)). In contrast, a recent study (Raymond et al., 2024 [↗](#)) reported that the population responses to conditioned and appetitive taste stimuli converge to a much greater degree than we observed after CTA extinction. One explanation for the difference between Raymond et al. and our findings is that we used two conditioning sessions with a novel tastant, while Raymond et al. used a single conditioning session with a familiar tastant. Thus, our paradigm likely elicited stronger CTA, and thus a more enduring ability to discriminate tastants. Interestingly, in contrast to the WT condition, extinction in the Shank3 KO leads to a rapid and stronger convergence of population taste-evoked responses in AIC, which ultimately reduces the ability to decode the identity of taste stimuli from their responses. Degraded sensory coding has been observed in sensory cortices of other ASD mouse models (Monday et al., 2023 [↗](#); Wang and Feldman, 2024 [↗](#)); thus, unstable and poorly differentiated cortical responsivity to sensory stimuli is a pathogenic feature shared across brain regions and gene mutations in ASD.

Our finding that taste representations remain discriminative in AIC throughout CTA extinction raises an important question: what taste information does AIC population activity actually encode? In the hippocampus, representations of odor “identity” remain stable independent of learning, while those encoding odor “values” are susceptible to extinction (Biane et al., 2023 [↗](#)). In AIC, ensemble activity encodes both the identity and palatability of taste stimuli, as well as related decision-making, in sequential temporal epochs (Katz et al., 2001 [↗](#); Sadacca et al., 2016 [↗](#); Livneh et al., 2017 [↗](#); Lavi et al., 2018 [↗](#); Chen et al., 2021 [↗](#); Staszko et al., 2022 [↗](#); Kogan and Fontanini, 2024 [↗](#); Raymond et al., 2024 [↗](#)). The population AIC activity we analyzed (3-second duration after delivery) is unlikely to simply encode taste identity, because if that were the case, the defined taste representations would be independent of learning and should be present even in animals that didn’t receive LiCl. Instead, the AIC population activity we measure likely encodes primarily taste palatability, or “valence.” This possibility would also explain the poor decoding performances of classifiers trained in the CS-only condition, where water and saccharin share similar innate palatability (Staszko et al., 2022 [↗](#)).

Flexible processing of new sensory information in the cortex enables animals to update their perceptions and modify behaviors that are critical for survival, and disruptions in this process could contribute to maladaptive behaviors in neurodevelopmental disorders. Here, we show that a higher signal-to-noise ratio in the AIC of Shank3 KO mice may hinder the updating of taste valence following negative experiences, leading to slower CTA learning and faster extinction. This rapid extinction is associated with rapid loss of taste discriminability within AIC, and this response instability ultimately compromises the ability to decode taste identity from AIC responses. More broadly, these data suggest that behavioral inflexibility in some forms of ASD could arise from a reduced ability to update behavior when the valence of a sensory input changes.

Materials and methods

Key Resources Table

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Chicken anti-GFP	Aves Labs	Cat#GFP-1020; RRID: AB_2307313
Mouse anti-NeuN, MAB-377	Millipore	Cat#MAB-377; RRID: AB_2298772
Goat anti-chicken Alexa-488	ThermoFisher Scientific	Cat#A-11039; RRID: AB_142924
Goat anti-mouse Alexa-594	ThermoFisher Scientific	Cat#A-11032; RRID: AB_2534091
Bacterial and virus strains		
AAV1-syn-jGCaMP8s	a gift from GENIE project ⁴⁴	Addgene viral prep: 162374-AAV1; RRID: Addgene_162374-AAV1
Chemicals, peptides, and recombinant proteins		
Saccharin sodium	Fisher Scientific	Cat#S394-500
Lithium chloride	ThermoFisher Scientific	Cat#A10531.22
Experimental models: Organisms/strains		
Mouse C57BL/6	The Jackson Laboratory	Strain#000664; RRID: IMSR_JAX:000664
Mouse Shank3B Δ exon 13-16	The Jackson Laboratory	Strain#017688; RRID: IMSR_JAX:017688
Software and algorithms		
CellReg	Sheintuch et al. ⁹⁸	
Suite2P	Pachitariu et al. ⁹⁶	https://github.com/cortex-lab/Suite2P ; RRID: SCR_016434
Python	Python	https://www.python.org ; RRID: SCR_008394
GraphPad Prism	GraphPad	http://www.graphpad.com/ ; RRID: SCR_002798
Other		
Gradient-index (GRIN) lens (OD: 0.5 mm, length: 3.91 $\pm$ 0.54 mm, working distance: 0.2 mm)	GoFoton	Cat#CLHS050GFT126

Resource Availability

Lead contact and materials availability

Further information and requests for resources and reagents should be directed to the lead contact, Gina G. Turrigiano (turrigiano@brandeis.edu [✉](#)).

Experimental Model and Subject Details

All experimental procedures were approved by Brandeis University Institutional Animal Care and Use Committee and followed the National Institute of Health guidelines for the Care and Use of Laboratory Animals. C57BL/6 and Shank3B knockout (Δ exon 13-16, JAX: 017688) mouse strains were obtained from the Jackson Laboratory, and the colonies were maintained in Foster Biomedical Research Labs at Brandeis University. Adult mice (8-16 weeks old) were used in experiments performed in this study. All animals were grouped with their littermates or housed individually once they entered the behavioral training paradigm. In all cases, animals were

housed in a humidity- and temperature-controlled environment and entrained to a 12-hour light-dark cycle (light phase from 7:00-7:00), with ad libitum access to food and water unless described otherwise. Mice of each sex were assigned equally to each experimental condition. In this paper, 7, 9, and 9 animals were assigned to the CS-only (WT), CTA (WT), and CTA (KO) groups in the two-conditioning behavior experiments (Figures 1 [↗](#) and 3 [↗](#)). For calcium imaging analysis, 1 and 2 animals were excluded from the CTA (WT) and CTA (KO) groups, respectively, due to a lack of sufficient GCaMP expression (Figures 1 [↗](#)-6 [↗](#)). In the separate single-conditioning paradigm, 13 and 13 animals were used in the CTA (WT) and CTA (KO) groups.

Viral Vectors

The AAV serotype 1 expressing pGP-AAV-syn-jGCaMP8s (Addgene: 162374) was obtained from Addgene.

Stereotaxic Surgery

Mice (8-10 weeks old) were anesthetized with a constant airflow containing 2% isoflurane throughout the procedure. The skull was exposed, and a craniotomy was created above the right AIC. AAV expressing GCaMP8s (titer: $\sim 1.2 \times 10^{13}$, 1:3 dilution, 800 nL in total) was slowly infused into AIC (anterior-posterior (AP) with reference to bregma: 1.0 mm, medial-lateral (ML): -3.8 mm, dorsal-ventral (DV): with reference to the brain surface: -2.0 mm) through a micropipette connected to a microinjector (Narishige, MO-10), at a rate of approximately 200 nL/min. To allow the virus particles to spread, the pipette was kept in place for another 5 minutes before being slowly withdrawn from the site. Two weeks after the virus injection, the mice (10-12 weeks old) then underwent the implantation of a gradient-index (GRIN) lens and a customized titanium head bar. Mice were anesthetized as described above, and a dose of dexamethasone (2 mg/kg) was administered subcutaneously at the beginning of the surgery to minimize brain swelling. Once the skull was revealed, a craniotomy slightly larger than 0.5 mm x 0.5 mm was made at the site of the previous virus injection. To create a leading track, a 25-gauge blunt needle was slowly lowered until it reached above AIC (AP: +1.0 mm, ML: -3.8 mm, DV: -1.8 mm), remained in place for five minutes and was withdrawn from the site. A GRIN lens (diameter: 0.5 mm, length: ~4 mm, CLHS050GFT126, GoFoton) was then gradually lowered through a customized holder until its bottom surface rested on top of AIC (AP: +1.0 mm, ML: -3.8 mm, DV: -2.0 mm). The lens was secured to the skull by Vetabond and the UV-curable adhesive, after which a titanium head-bar was installed. Metabond tinged with black ink was applied to further cement the lens and the head bar while covering and sealing the skull. After the operation, animals were kept and monitored individually in their home cages, and a daily dose of meloxicam (1-2 mg/kg) was given subcutaneously for at least three days to reduce brain inflammation. Animals were allowed for at least one week of recovery before starting behavioral training.

Head-restrained CTA Learning Paradigm

One day before the mice (13-15 weeks old) began training in the head-fixed position, ad libitum water access was removed from their home cages to motivate their engagement in the subsequent tasks. The restricted water access was then given daily through the motorized spout during the training task or in the home cage for thirty minutes (described in detail below). Each training day, animals were placed on a customized wheel (Livneh et al., 2017 [↗](#)) with their heads restrained and underwent a thirty-minute training session consisting of 30-35 trials. In each trial, a two-second monotonal acoustic cue (2 kHz) was initiated to signal the onset of the trial, with a motorized spout simultaneously moving to the location close to the animal's mouth. Mice then learned to perform five dry licks in order to receive the liquid through the spout. The gravity-based liquid delivery, once initiated, lasted 3 seconds, after which the spout was moved away from the animal. A 10-second intertrial rest followed before the next trial started. A trial was considered failed if the animal did not reach the threshold to initiate the liquid delivery within sixty seconds after the spout was in place.

Habituation

During the habituation stage, water was used as the tastant. If animals refused to lick the spout throughout the session or their body weight dropped over 15 percent of the initial weight, an additional ad libitum access to water was granted for 30 minutes in the home cage. On average, it took 10 days for the animals to become familiarized with the task, reaching consistent performance without failed trials within thirty minutes. By the end of the habituation, the spout became the only daily source of liquid the animals received, and their body weight stabilized.

Conditioning

Once habituated, animals then received two conditioning sessions to induce the formation of conditioned taste aversion (CTA). On each conditioning day, animals underwent the training session with the same scheme as in habituation, except that saccharin (25 mM, CS) was given instead of water. Five minutes after the session, a dose of lithium chloride (LiCl, 0.15 mM, 2% body weight, US) was intraperitoneally injected to induce gastric malaise. Afterward, animals remained in place for another 25 minutes before returning to the home cage. To overcome the potential association between malaise and the context, which might confound animals' subsequent performance, animals were allowed to receive water in a recovery session (in the same environment) the next day after each conditioning; during this time, animals usually resumed their engagement in the task.

CTA retrieval and extinction testing

After the second recovery session, we subjected animals to consecutive test sessions for five days to measure the retrieval and extinction of CTA memory. In each test session, water was given for the first three trials to engage animals in the task. Afterward, water and saccharin were delivered in a semi-randomized order such that animals could not predict the identity of the tastant in the upcoming trial while overall receiving similar numbers of trials that delivered water and saccharin.

Single-conditioning CTA paradigm

A modified taste learning paradigm was conducted to examine whether Shank3 knockout mice expressed taste aversion when presented with choices after one conditioning. After the first conditioning session and the subsequent recovery session, animals were immediately subjected to a retrieval test session in which water and saccharin at various concentrations (25 mM, 10 mM, 5 mM, 1 mM) were delivered in a semi-randomized order. Trial-averaged licks and lick ratios were quantified as described below.

Behavior Data Analysis

Trial-averaged lick calculation

Throughout the paradigm, licks were detected by a capacitive sensor (MPR121, Adafruit) attached to the spout, which generated TTL signals recorded with the customized code and time-stamped to the frames during two-photon image acquisition. To obtain trial-averaged licks of a tastant (Figure 1 and 3), total lick counts during the 3-second delivery were averaged across trials delivering that tastant in a given session. In the cases where the licks of saccharin and water were compared (Figure 3), the ratio was further calculated by dividing the trial-averaged licks of saccharin over that of water.

Logistic curve fitting

A logistic curve was fitted to the trial-averaged lick ratios as a function of time (session) to describe the temporal dynamics of the licking behavior (Kogan and Fontanini, 2024). The logistic function was fitted using the `curve_fit` function in the Python `scipy` model and is defined as:

$$f(t) = \frac{L}{1 + e^{-k(t-t_0)}} \quad (1)$$

where $f(t)$ is the fitted lick ratio at time t ; L is the maximum lick ratio; k denotes the growth rate, and t_0 indicates the inflection point. The derivative of the logistic function was calculated as:

$$f'(t) = \frac{Lke^{-k(t-t_0)}}{(1 + e^{-k(t-t_0)})^2} \quad (2)$$

Immunohistochemistry

Animals were deeply anesthetized with isoflurane and perfused with 4% paraformaldehyde (PFA). The whole brain was extracted, post-fixed in 4% PFA for 24 hours, and sliced on a vibrating microtome (Leica Vibratome VT 1200s). Coronal brain slices (50 μ m) containing AIC were collected; to verify the placement of the GRIN lens, the slices with the visible track left by the lens were selected, washed three times with PBS, permeabilized with the blocking buffer (5% goat serum/3% BSA/0.3% Triton X-100 in PBS) at room temperature for 2 hours, and incubated with primary antibodies diluted in the blocking buffer at 4°C overnight. To confirm the expression of GCaMP8s, chicken anti-GFP (1:1000, Aves Labs) and mouse anti-NeuN (1:500, MAB-377, Millipore) were used. On the next day, the slices were washed three times with PBS and incubated with secondary antibodies diluted in the buffer containing 5% goat serum and 3% BSA at room temperature for 3 hours (goat anti-chicken Alexa-488 and goat anti-mouse Alexa-594, 1:400, Thermo-Fisher). After another three washes with PBS, the slices were mounted on the slides using the DAPI-Fluoromount-G mounting medium (SouthernBiotech), cover-slipped, and stored at 4°C until imaging.

Confocal Imaging

To verify the expression of GCaMP8s and the placement of the GRIN lens, stained slices were imaged on a laser scanning confocal microscope (Zeiss LSM880) using the tiling scan under a 20x objective, with a frame size of 1024 pixels x 1024 pixels. For each hemisphere, the images were stitched online using the ZEN Black software (Zeiss) and then analyzed using ImageJ/FIJI (NIH, US).

Two-photon Imaging

Acquisition

Two-photon imaging was conducted using a resonant-scanning two-photon microscope (Neurolabware, 31 frames per second; 512 pixels x 512 pixels) with a 10X objective (Thorlabs, NA: 0.50, air, working distance: 7.77 mm, zoom factor: 2.8). GCaMP8s was excited at 920 nm using a tunable ultrafast laser (SpectraPhysics Insight X3, 50-80 mW measured at the front aperture of the objective). Fields of view of all imaging were selected approximately 200 μ m below the bottom surface of the GRIN lens. On average, 750 frames (25 seconds) were obtained for each trial.

Processing and registration

For each imaging session, frames from all successful trials were included and concatenated into a series. In each movie, the movement was corrected, the regions of interest (ROI) corresponding to individual neurons in the field of view were identified, and the raw fluorescence from the cell bodies and neuropils was extracted using Suite2P (Pachitariu et al., 2017 [DOI](#)). The processed images were then inspected, and if needed, the ROIs were manually selected or excluded using the Suite2P GUI so that only the cell bodies were used for further analysis. Only the ROIs near the center of the field of view were included to avoid any potential peripheral distortion of fluorescence by spherical aberration (Sattin et al., 2024 [DOI](#)). The time series of neuropil-corrected calcium signals of individual neurons were obtained by subtracting the time series of neuropil fluorescence (scaled by 0.7) from that of the raw fluorescence of cell bodies. In the case where multi-day cell registration was required (i.e., identification of the same cells across sessions, [Figure 4](#) [DOI](#) and [6](#) [DOI](#)), Spatial footprints of each neuron were extracted from the field of view of each session with in-house customized code and were aligned across sessions using the probabilistic modeling method implemented in CellReg (Sheintuch et al., 2017 [DOI](#)). Centroid distance was recommended by CellReg in most cases, and a probability threshold of 0.5 was used.

Cell inclusion criteria

Cells were identified by the classifier built and trained in Suite2p. Once identified, each cell's extracted somatic and neuropil calcium traces were then inspected; if the contamination from neuropils was severe (i.e., the magnitude of neuropil calcium signals was consistently higher than that of somatic signals), that cell was excluded from the analysis.

Calcium Imaging Data Analysis

Calculation of $\Delta F/F_0$ and stimulus responsiveness

Z-scored changes in calcium signals were used to represent the temporal activity dynamics of individual neurons in all analyses. In each trial, neuropil-corrected calcium traces of the same duration (total 656 frames) were first extracted, which consisted of 155 frames of baseline (5 seconds before the cue onset), 62 frames of the acoustic cue (2 seconds), 15 frames of dry licks (~0.5 seconds), 112 frames of tastant delivery (3.5 seconds), and 312 frames of post-delivery rest (10 seconds). On average, it took animals no more than 0.5 seconds (15 frames) to reach the threshold of five dry licks; however, during the early stages of memory extinction, animals occasionally hesitated or even failed to participate in some trials, prolonging the dry-lick period to varying degrees. In such cases, we only included 15 frames starting from the initiation of the first dry lick to ensure that the lengths of the time series were consistent across animals, trials, and sessions. For each neuron in a given trial, we first computed its changes in fluorescence ($\Delta F/F_0$) by subtracting the mean fluorescence of baseline (F_0) from the fluorescence at each time point and dividing the result values by F_0 . Next, we calculated the z-scores using the formula $(F_{(t)} - F_{\text{baseline}}) / \text{SD}$, where $F_{(t)}$ represents the $\Delta F/F_0$ at time t , and F_{baseline} and SD denote the mean and standard deviation of the $\Delta F/F_0$ values during the baseline. To determine whether a neuron is responsive to a taste stimulus in a given session (Figure 1, 2, 4, 5), we compared the mean z-scored activity during the taste delivery (last 2 seconds) to the mean z-scored baseline activity (2 seconds before the cue onset) across trials using paired t-test. A neuron was categorized as “activated” and “suppressed” during the tastant delivery if there was a significant increase and decrease in activity compared to the baseline, respectively. We omitted the first second of the tastant delivery when calculating the mean activity to avoid a potential carryover from the dry lick responses (Chen et al., 2021). In the analyses that used the amplitude of the trial-averaged response (z-scored $\Delta F/F_0$, Figure 1, 2, 4, 5, 6), we first averaged the z-scored calcium traces across trials and then calculated the values of the mean activity during tastant deliveries as described above.

Coactivity analysis

The calculation of taste-elicited coactivity patterns between neuronal pairs was adapted from Twarkowski et al., 2022. Within each session, trials of the same tastant deliveries were selected, and the Pearson correlation coefficient between two neurons' mean z-scored $\Delta F/F_0$ values during the tastant delivery across these trials was calculated. To determine the stimulus-independent baseline coactivity, the mean z-scored $\Delta F/F_0$ values during the 2-second period before the cue onset were used to compute the Pearson correlation coefficient.

Pairwise signal and noise correlations

The signal and noise correlations were calculated as previously described (Marks and Goard, 2021). The signal correlation was obtained in a given session by calculating the Pearson correlation between two neurons' full trial-averaged activity traces during the tastant delivery. To estimate the correlated variability (noise), we first subtracted a neuron's trial-averaged activity trace from its z-scored activity trace in each trial. The subtracted traces (residuals) from all trials within a session were then concatenated into a whole series representing trial-to-trial variability in the neuron's activities. The Pearson correlation coefficient was calculated between two neurons' concatenated, subtracted traces to represent correlated variability.

Single-cell discriminability index

The discriminability index was adapted from the previously reported method (Kong et al., 2023) and indicated a neuron's response preference between taste stimuli (water or saccharin). In a given session, the discriminability index for each neuron is defined as:

$$\text{Discriminability index} = \frac{\mu_1 - \mu_2}{\sqrt{\frac{1}{2}(\sigma_1^2 + \sigma_2^2)}} \quad (3)$$

where μ_1 and μ_2 denote the trial-averaged activity amplitude averaged across saccharin and water trials, respectively, and the denominator is the pooled standard deviation. A positive index indicates that the neuron responds more preferentially to saccharin, while a more negative index indicates a higher response preference for water.

Euclidean distance

We calculated the pairwise Euclidean distance to estimate the similarity between population responses to saccharin and water (Levitan et al., 2019; Chen et al., 2021; Raymond et al., 2024). In a given test session, for each animal, we constructed the trial-averaged ΔF values of all neurons into two population vectors, one responding to saccharin and the other to water. The Euclidean distance between these two vectors is then defined as:

$$\text{Normalized Euclidean distance} = \frac{\sqrt{\sum_{i=1}^n (v_{1i} - v_{2i})^2}}{\sqrt{n}} \quad (4)$$

Where v_1 and v_2 denote the population vector in response to saccharin and water, respectively, and n is the number of neurons included in the vectors. To mitigate the influence on Euclidean distance by the number of neurons included in the calculation (i.e., the dimension), we normalized the distance by the square root of the dimension (Levitan et al., 2019).

Decoding analysis

We trained a support vector classifier for each animal to predict the tastant class (water or saccharin) in a given session, using the SVC function with a linear kernel in the Python scikit-learn module (Woods et al., 2020; Twarkowski et al., 2022; Raymond et al., 2024). We first constructed population response vectors for each trial (labeled as saccharin or water), and these vectors constituted the dataset for model training and testing. The linear decoder is defined as:

$$L(t) = \theta(w^T v(t) + b) \quad (5)$$

where $L(t)$ is the predicted tastant class, and $v(t)$ is the population response vector for the trial t . $\theta(z)$ is the Heaviside step function that ensures hard classification, while w denotes the vector that contains weights assigned to each neuron, and b is the bias term (Bishop, 2006). We then used 10-fold cross-validation, a standard method for model validation and evaluation, to train and test the data. Cross-validation produced an averaged accuracy score that indicated the classifier's performance. To calculate chance-level prediction, we shuffled the labels and trained the classifier as described above for 1000 iterations. This shuffling process resulted in an average 50-percent accuracy score by chance. To test the stability of insular taste representations, we used the classifier trained on the data in test 1 to predict the tastant classes in the subsequent sessions and calculated the accuracy score accordingly.

Quantification and Statistical Analysis

All statistical analyses were performed using the Python scipy, statsmodels, or pingouin packages. To assess the main effects and interactions in experiments comparing the same animals or neurons across sessions under different conditions, a paired t-test or repeated-measures mixed ANOVA was used unless stated otherwise. For experiments requiring specific comparisons between conditions, a linear mixed-effect model was fitted to the data, and the relevant coefficient

was extracted. If the comparison could not be directly inferred from the coefficient, a post hoc t-test was performed. The significance levels were labeled by asterisks (*): $P < 0.05$; (**): $P < 0.01$; ***: $P < 0.001$; ****: $P < 0.0001$.

Supplementary figure legends

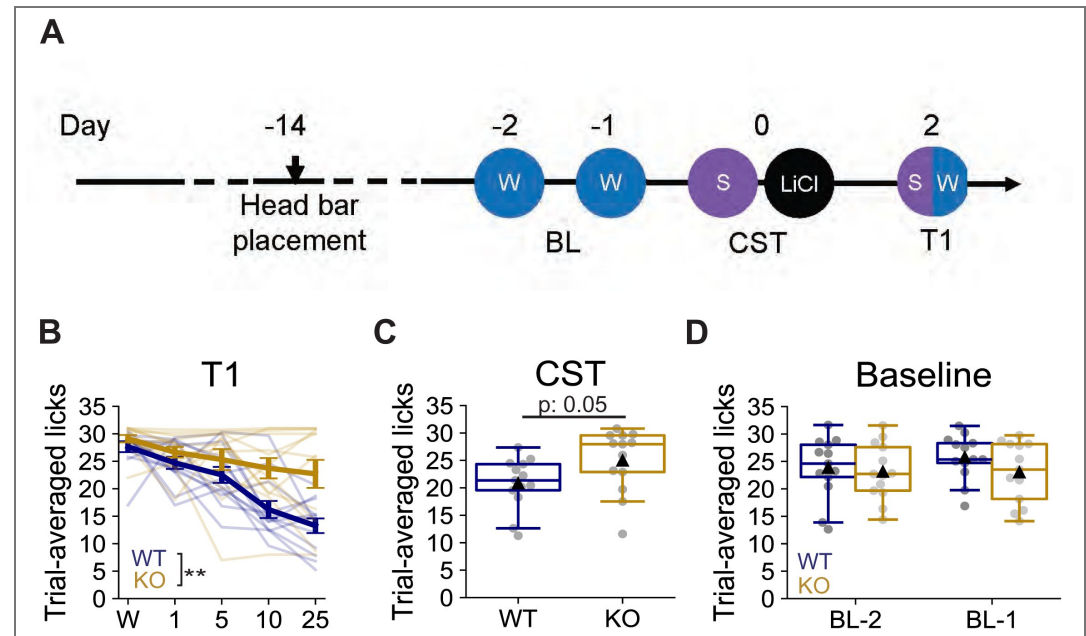


Figure 1 - Figure Supplement 1. Shank3 knockout hinders CTA acquisition in a single-conditioning paradigm

(A) The modified CTA learning paradigm. CS: conditioned stimulus, 25 mM saccharin, US: unconditioned stimulus, lithium chloride (LiCl, 0.15M, 2% body weight). BL: baseline, CST: conditioning session, T: retrieval test session, S: saccharin, W: water. **(B)** Trial-averaged licks of water and varying concentrations of saccharin (25 mM, 10 mM, 5 mM, 1 mM) in the retrieval test session (repeated-measures mixed ANOVA, genotype, 0.0049, taste, $p < 0.0001$, interaction, $p = 0.0033$). **(C)** Trial-averaged saccharin licks during the conditioning session (unpaired t-test, $p = 0.0547$). **(D)** Trial-averaged water licks during the two baseline sessions (WT: $n = 13$ animals, KO: $n = 13$ animals, BL-2, unpaired t-test, $p = 0.7402$; BL-1, unpaired t-test, $p = 0.1662$).

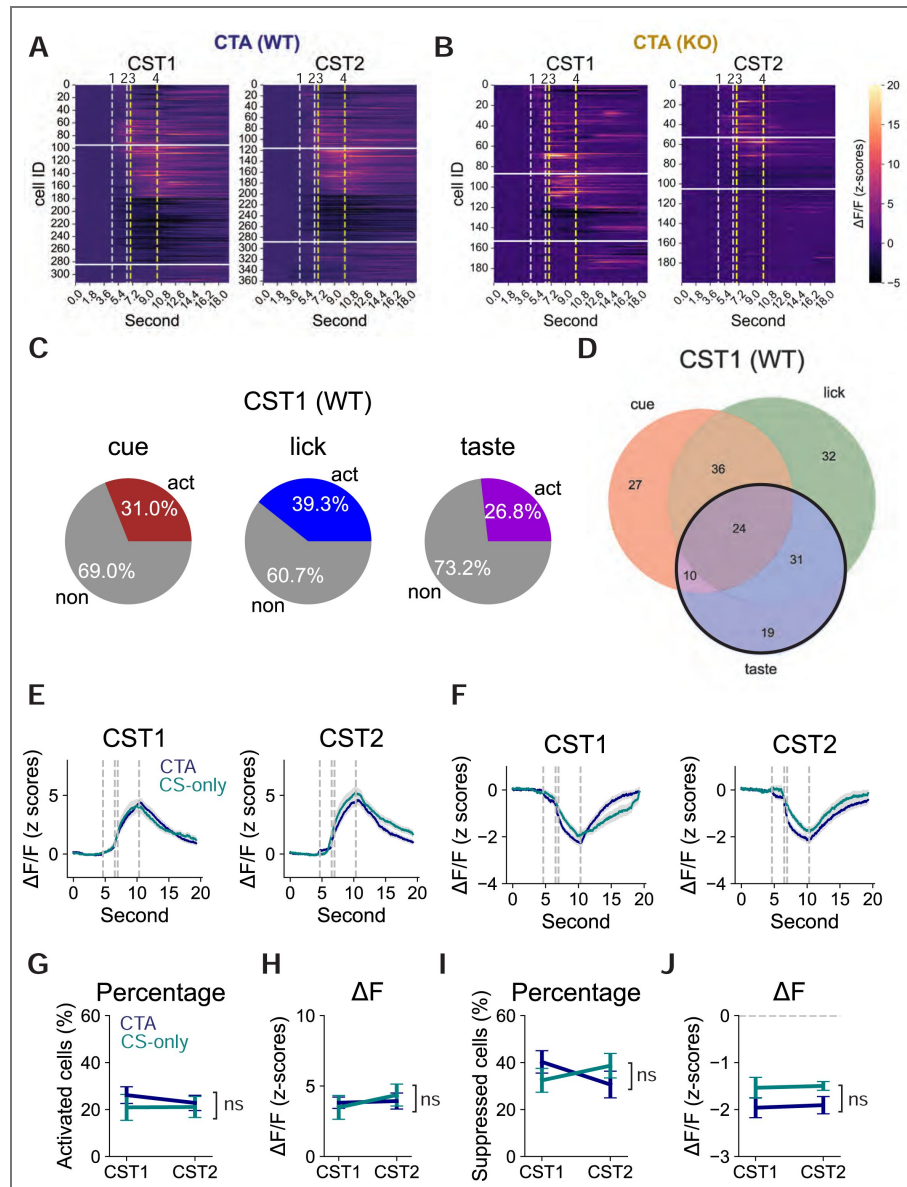


Figure 1 – Figure Supplement 2. Comparison of taste-elicited AIC neuronal responses in the CTA and CS-only groups during conditioning

(A, B) Heatmaps showing z-scored changes in calcium responses ($\Delta F/F$) from all recorded neurons during conditioning in WT (A, CST1: 313 cells, CST2: 363 cells) and KO (B, CST1: 194 cells, CST2: 200 cells) groups. 1: cue onset; 2: spout in place; 3: tastant delivery starts; 4: tastant delivery ends. The 3-second taste delivery is marked by yellow dashed lines. In each heatmap, cells are sorted in descending order based on the onset of their evoked responses. Cells that were responsive to taste delivery (between white horizontal lines) were selected and shown in the main figures 1H and 1I. (C) Percentage of neurons activated by the cue (left), lick movement (middle), and the tastant (right) in the CTA (WT) group (313 cells) during the first conditioning session (CST1). (D) Venn diagram showing the overlaps among cue-, lick-, and tastant-responsive neurons in (C). (E) Calcium traces averaged across activated neurons in the CTA (WT) and CS-only (WT) groups during CST1 (left) and CST2 (right). (F) Calcium traces averaged across suppressed neurons in the CTA (WT) and CS-only (WT) groups during CST1 (left) and CST2 (right). (G) Change in the percentage of activated neurons from CST1 to CST2 (CTA vs. CS-only, repeated-measures mixed ANOVA, group: $p = 0.5202$, session: $p = 0.5460$, interaction: 0.5372). (H) Change in the calcium fluorescence ($\Delta F/F$) of activated neurons from CST1 to CST2 (CTA vs. CS-only, repeated-measures mixed ANOVA, group: $p = 0.8465$, session: $p = 0.1287$, interaction: 0.1650). (I) Change in the percentage of suppressed neurons from CST1 to CST2 (CTA vs. CS-only, repeated-measures mixed ANOVA, group: $p = 0.9889$, session: $p = 0.5189$, interaction: 0.0345). (J) Change in the calcium fluorescence ($\Delta F/F$) of suppressed neurons from CST1 to CST2. Dashed lines mark the value 0 (CTA vs. CS-only, repeated-measures mixed ANOVA, group: $p = 0.1166$, session: $p = 0.6204$, interaction: 0.9379).

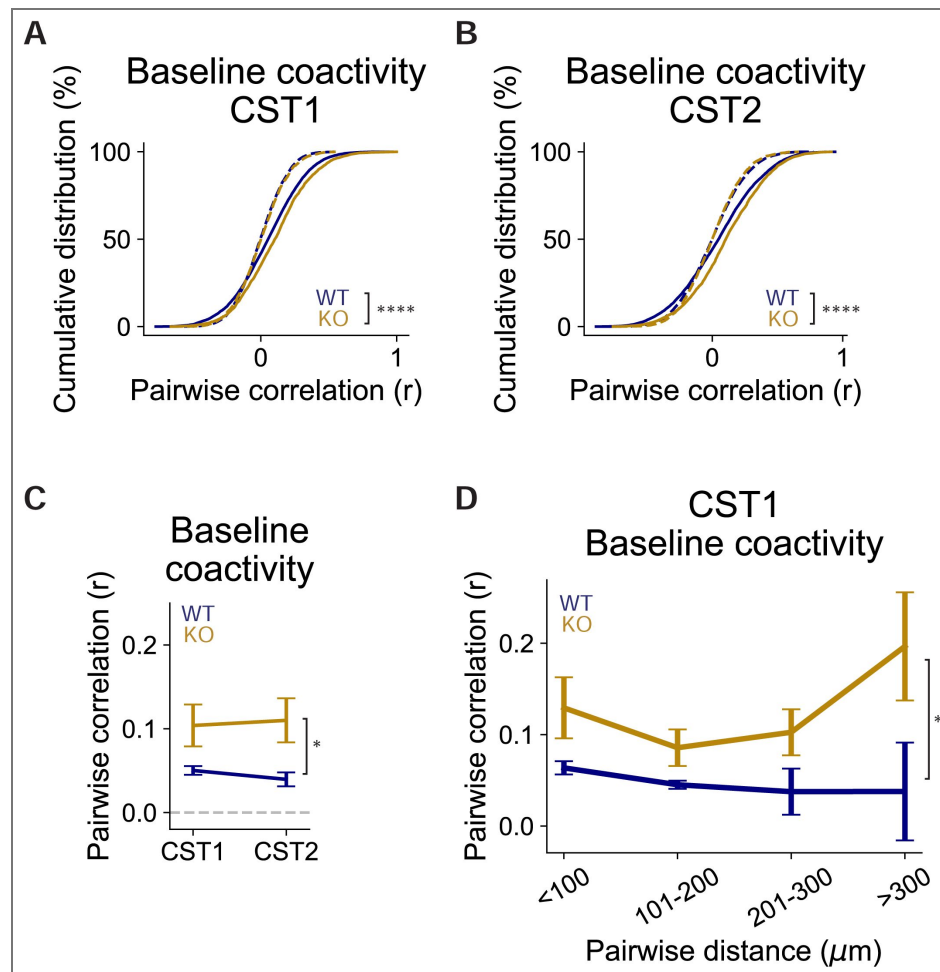


Figure 2 – Figure Supplement 1. Shank3 knockout increases baseline coactivity in AIC during CTA acquisition

(A, B) Cumulative distribution of baseline coactivity (Pearson coefficients) between neurons during CST1 (A) and CST2 (B). Dashed lines indicate the shuffled data (CST1: WT, $n = 6201$ pairs, KO, $n = 2645$ pairs, WT vs. KO, Kolmogorov-Smirnov test, $p < 0.0001$; CST2: WT, $n = 8476$ pairs, KO, $n = 2857$ pairs, WT vs. KO, Kolmogorov-Smirnov test, $p < 0.0001$). **(C)** Change in the average baseline coactivity from CST1 to CST2, plotted by animal. The dashed line indicates the chance-level correlation. (WT vs. KO, repeated-measures mixed ANOVA, genotype, $p = 0.0225$, session, $p = 0.7014$, interaction: $p = 0.2628$). **(D)** Average baseline coactivity in CST1, grouped by inter-neuronal distance (WT vs. KO, repeated-measures mixed ANOVA, genotype, $p = 0.0364$, distance, $p = 0.2776$, interaction: $p = 0.1821$).

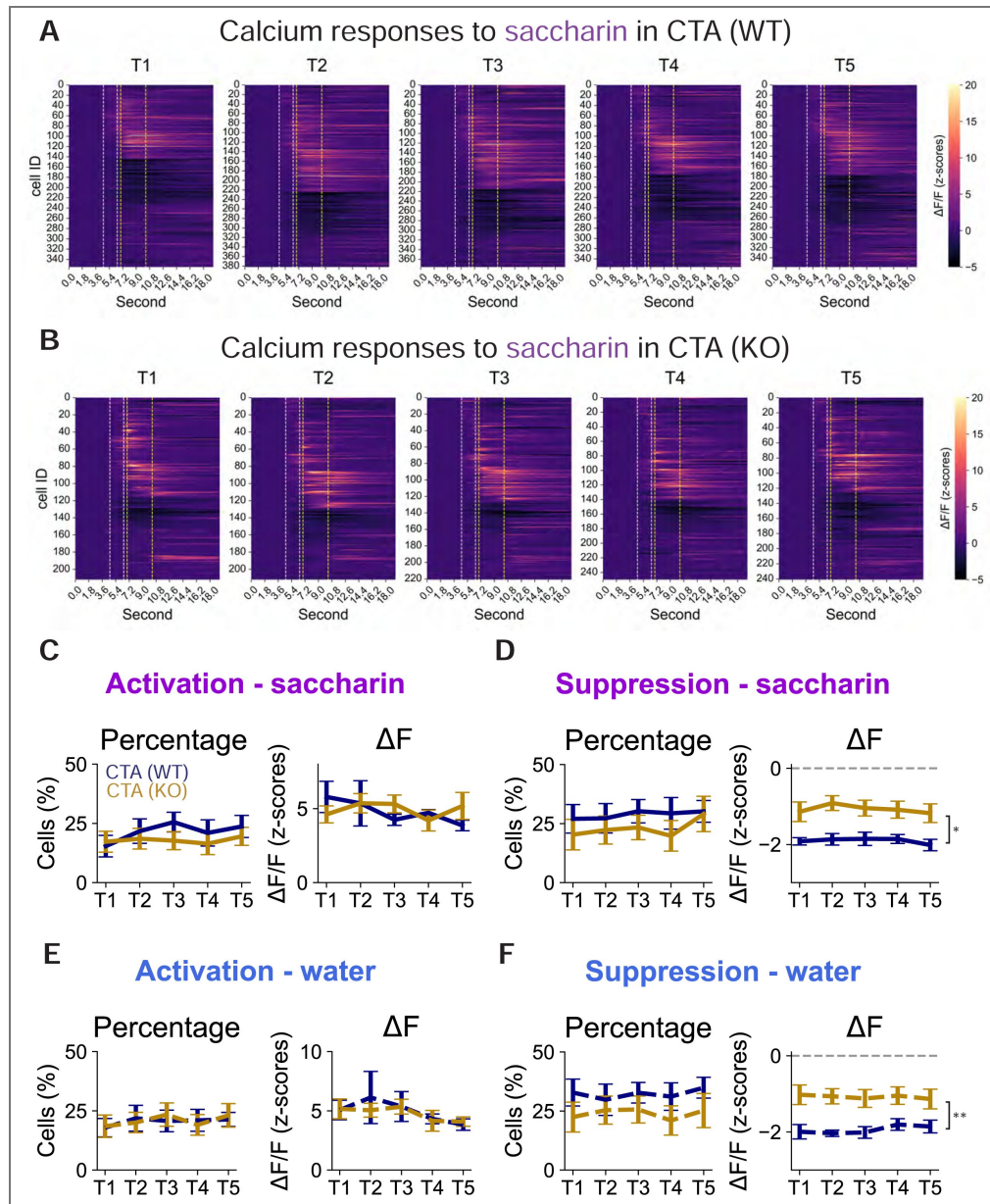


Figure 5 – Figure Supplement 1. The number and amplitude of AIC neuronal responses to taste remain stable during CTA extinction

(**A**) Heatmaps showing z-scored changes in calcium responses ($\Delta F/F$) to saccharin from all recorded neurons in the CTA (WT) group throughout retrieval test sessions (T1: 355 cells, T2: 382 cells, T3: 376 cells, T4: 356 cells, T5: 358 cells). (**B**) Heatmaps showing z-scored changes in calcium responses ($\Delta F/F$) to saccharin from all recorded neurons in the CTA (KO) group throughout retrieval test sessions (T1: 212 cells, T2: 212 cells, T3: 221 cells, T4: 250 cells, T5: 241 cells). (**C**) The percentage (left) and the $\Delta F/F$ (right) of neurons activated by saccharin from T1 to T5 (percentage: WT vs. KO, repeated-measures mixed ANOVA, genotype, $p = 0.5560$, session, $p = 0.0595$, interaction, $p = 0.2615$; $\Delta F/F$: repeated-measures mixed ANOVA, genotype, $p = 0.9601$, session, $p = 0.5807$, interaction, $p = 0.4845$). (**D**) The percentage (left) and the $\Delta F/F$ (right) of neurons suppressed by saccharin from T1 to T5 (percentage: WT vs. KO, repeated-measures mixed ANOVA, genotype, $p = 0.4674$, session, $p = 0.2587$, interaction, $p = 0.6529$; $\Delta F/F$: repeated-measures mixed ANOVA, genotype, $p = 0.0201$, session, $p = 0.0998$, interaction, $p = 0.8582$). (**E**) The percentage (left) and the $\Delta F/F$ (right) of neurons activated by water from T1 to T5 (percentage: WT vs. KO, repeated-measures mixed ANOVA, genotype, $p = 0.9612$, session, $p = 0.5387$, interaction, $p = 0.8687$; $\Delta F/F$: repeated-measures mixed ANOVA, genotype, $p = 0.7128$, session, $p = 0.2653$, interaction, $p = 0.8600$). (**F**) The percentage (left) and the $\Delta F/F$ (right) of neurons suppressed by water from T1 to T5 (percentage: WT vs. KO, repeated-measures mixed ANOVA, genotype, $p = 0.2913$, session, $p = 0.6114$, interaction, $p = 0.7659$; $\Delta F/F$: repeated-measures mixed ANOVA, genotype, $p = 0.0088$, session, $p = 0.7359$, interaction, $p = 0.3925$).

Acknowledgements

We thank Donald B. Katz and members of the Katz lab, and Stephen D. Van Hooser, for helpful comments and conceptual guidance. We also thank Dr. Mark L. Andermann for providing technical advice and the 3D printing template for the running wheel. This work was supported by R35NS111562 (GGT).

Additional information

Data and code availability

All data generated in this study are included in the figures and supplemental figures. Custom Python codes used for analyses of calcium imaging data can be found at https://github.com/geonwu-neuro/Wu_2025.

Funding

Funder	Grant reference number
National Institute of Neurological Disorders and Stroke	R35NS111562

Author ORCID iDs

Chi-Hong Wu: <http://orcid.org/0000-0001-6391-0747>

Gina G Turrigiano: <http://orcid.org/0000-0002-4476-4059>

References

- Arieli E**, Younis N, Moran A (2022) Distinct Progressions of Neuronal Activity Changes Underlie the Formation and Consolidation of a Gustatory Associative Memory. *J Neurosci* **42**:909-921
- Avery JA**, Ingeholm JE, Wohltjen S, Collins M, Riddell CD, Gotts SJ, Kenworthy L, Wallace GL, Simmons WK, Martin A (2018) Neural correlates of taste reactivity in autism spectrum disorder. *NeuroImage: Clinical* **19**:38-46
- Balasco L**, Pagani M, Pangrazzi L, Chelini G, Ciancone Chama AG, Shlosman E, Mattioni L, Galbusera A, Iurilli G, Provenzano G, *et al.* (2022) Abnormal whisker-dependent behaviors and altered cortico-hippocampal connectivity in *Shank3b* $-/-$ mice. *Cerebral Cortex* **32**:3042-3056
- Banerjee A**, Parente G, Teutsch J, Lewis C, Voigt FF, Helmchen F (2020) Value-guided remapping of sensory cortex by lateral orbitofrontal cortex. *Nature* **585**:245-250
- Banerjee A**, Rikhye RV, Breton-Provencher V, Tang X, Li C, Li K, Runyan CA, Fu Z, Jaenisch R, Sur M (2016) Jointly reduced inhibition and excitation underlies circuit-wide changes in cortical processing in Rett syndrome. *Proc Natl Acad Sci USA*. Accessed February 5, 2025
<https://pnas.org/doi/full/10.1073/pnas.1615330113>
- Bariselli S**, Contestabile A, Tzanoulina S, Musardo S, Bellone C (2018) SHANK3 Downregulation in the Ventral Tegmental Area Accelerates the Extinction of Contextual Associations Induced by Juvenile Non-familiar Conspecific Interaction. *Front Mol Neurosci* **11**:360
- Bartolo R**, Saunders RC, Mitz AR, Averbach BB (2020) Information-Limiting Correlations in Large Neural Populations. *J Neurosci* **40**:1668-1678
- Berman DE**, Dudai Y (2001) Memory extinction, learning anew, and learning the new: dissociations in the molecular machinery of learning in cortex. *Science* **291**:2417-2419
- Biane JS**, Ladow MA, Stefanini F, Boddu SP, Fan A, Hassan S, Dundar N, Apodaca-Montano DL, Zhou LZ, Fayner V, *et al.* (2023) Neural dynamics underlying associative learning in the dorsal and ventral hippocampus. *Nat Neurosci* **26**:798-809
- Bishop CM** (2006) *Pattern recognition and machine learning* New York: Springer.

- Bozdagi O**, Sakurai T, Papapetrou D, Wang X, Dickstein DL, Takahashi N, Kajiura Y, Yang M, Katz AM, Scattoni ML, *et al.* (2010) Haploinsufficiency of the autism-associated Shank3 gene leads to deficits in synaptic function, social interaction, and social communication. *Molecular Autism* **1**:15
- Chen K**, Kogan JF, Fontanini A (2021) Spatially Distributed Representation of Taste Quality in the Gustatory Insular Cortex of Behaving Mice. *Curr Biol* **31**:450
- Chen Q**, Deister CA, Gao X, Guo B, Lynn-Jones T, Chen N, Wells MF, Liu R, Goard MJ, Dimidschstein J, *et al.* (2020) Dysfunction of cortical GABAergic neurons leads to sensory hyper-reactivity in a Shank3 mouse model of ASD. *Nat Neurosci* **23**:520-532
- Chéreau R**, Bawa T, Fodoulou L, Carleton A, Pagès S, Holtmaat A (2020) Dynamic perceptual feature selectivity in primary somatosensory cortex upon reversal learning. *Nat Commun* **11**:3245
- Choi S-Y** (2018) Synaptic and circuit development of the primary sensory cortex. *Exp Mol Med* **50**:1-9
- Copping NA**, Berg EL, Foley GM, Schaffler MD, Onaga BL, Buscher N, Silverman JL, Yang M (2017) Touchscreen Learning Deficits and Normal Social Approach Behavior in the Shank3B Model of Phelan-Mcdermid Syndrome and Autism. *Neuroscience* **345**:155-165
- De la Casa G**, Lubow RE (1995) Latent inhibition in conditioned taste aversion: the roles of stimulus frequency and duration and the amount of fluid ingested during preexposure. *Neurobiol Learn Mem* **64**:125-132
- Deitch D**, Rubin A, Ziv Y (2021) Representational drift in the mouse visual cortex. *Current Biology* **31**:4327-4339.
- Delling JP**, Boeckers TM (2021) Comparison of SHANK3 deficiency in animal models: phenotypes, treatment strategies, and translational implications. *J Neurodevelop Disord* **13**:55
- Deng H**, Xiao X, Yang T, Ritola K, Hantman A, Li Y, Huang ZJ, Li B (2021) A genetically defined insula-brainstem circuit selectively controls motivational vigor. *Cell* **184**:6344-6360.
- Escobar ML**, Chao V, Bermúdez-Rattoni F (1998) In vivo long-term potentiation in the insular cortex: NMDA receptor dependence. *Brain Research* **779**:314-319
- Fletcher ML**, Ogg MC, Lu L, Ogg RJ, Boughter JD (2017) Overlapping Representation of Primary Tastes in a Defined Region of the Gustatory Cortex. *J Neurosci* **37**:7595-7605
- Garcia J**, Kimeldorf DJ, Koelling RA (1955) Conditioned aversion to saccharin resulting from exposure to gamma radiation. *Science* **122**:157-158
- Gehrlach DA**, Dolensek N, Klein AS, Roy Chowdhury R, Matthys A, Junghänel M, Gaitanos TN, Podgornik A, Black TD, Reddy Vaka N, *et al.* (2019) Aversive state processing in the posterior insular cortex. *Nat Neurosci* **22**:1424-1437
- Gil-Lievana E**, Ramírez-Mejía G, Urrego-Morales O, Luis-Islas J, Gutierrez R, Bermúdez-Rattoni F (2022) Photostimulation of Ventral Tegmental Area-Insular Cortex Dopaminergic Inputs Enhances the Salience to Consolidate Aversive Taste Recognition Memory via D1-Like Receptors. *Front Cell Neurosci* **16**:823220
- Glas A**, Hübener M, Bonhoeffer T, Goltstein PM (2021) Spaced training enhances memory and prefrontal ensemble stability in mice. *Current Biology* **31**:4052-4061.
- Gogolla N** (2017) The insular cortex. *Current Biology* **27**:R580-R586
- Gogolla N**, Takesian AE, Feng G, Fagiolini M, Hensch TK (2014) Sensory integration in mouse insular cortex reflects GABA circuit maturation. *Neuron* **83**:894-905
- Goldschlager J**, Cintron C, Hall R, Shields T, Tolbert GL, Woldebirhan R, Agarwal K, Joseph PV (2025) Taste processing in autism spectrum disorder: A translational scoping review. *Neuroscience & Biobehavioral Reviews* **170**:106031
- Haley MS**, Bruno S, Fontanini A, Maffei A (2020) LTD at amygdalocortical synapses as a novel mechanism for hedonic learning. *eLife* **9**
- Harris KD**, Mrsic-Flogel TD (2013) Cortical connectivity and sensory coding. *Nature* **503**:51-58

- Hazon O**, Minces VH, Tomàs DP, Ganguli S, Schnitzer MJ, Jercog PE (2022) Noise correlations in neural ensemble activity limit the accuracy of hippocampal spatial representations. *Nat Commun* **13**:4276
- Kafashan M**, Jaffe AW, Chettih SN, Nogueira R, Arandia-Romero I, Harvey CD, Moreno-Bote R, Drugowitsch J (2021) Scaling of sensory information in large neural populations shows signatures of information-limiting correlations. *Nat Commun* **12**:473
- Katz DB**, Simon SA, Nicolelis MAL (2001) Dynamic and Multimodal Responses of Gustatory Cortical Neurons in Awake Rats. *J Neurosci* **21**:4478-4489
- Katz DB**, Simon SA, Nicolelis MAL (2002) Taste-specific neuronal ensembles in the gustatory cortex of awake rats. *J Neurosci* **22**:1850-1857
- Khan AG**, Poort J, Chadwick A, Blot A, Sahani M, Mrcic-Flogel TD, Hofer SB (2018) Distinct learning-induced changes in stimulus selectivity and interactions of GABAergic interneuron classes in visual cortex. *Nat Neurosci* **21**:851-859
- Khoury CF**, Fala NG, Runyan CA (2022) The spatial scale of somatostatin subnetworks increases from sensory to association cortex. *Cell Reports* **40**:111319
- Kloth AD**, Badura A, Li A, Cherskov A, Connolly SG, Giovannucci A, Bangash MA, Grasselli G, Peñagarikano O, Piochon C, *et al.* (2015) Cerebellar associative sensory learning defects in five mouse autism models. *eLife* **4**:e06085 <https://doi.org/10.7554/eLife.06085>
- Kogan JF**, Fontanini A (2024) Learning enhances representations of taste-guided decisions in the mouse gustatory insular cortex. *Current Biology* **34**:1880-1892.
- Chandran S Kolatt**, Yiannakas A, Kayyal H, Salalha R, Cruciani F, Mizrahi L, Khamaisy M, Stern S, Rosenblum K (2023) Intrinsic Excitability in Layer IV–VI Anterior Insula to Basolateral Amygdala Projection Neurons Correlates with the Confidence of Taste Valence Encoding. *eNeuro* **10**:ENEURO.0302-22.2022
- Kong E**, Lee K-H, Do J, Kim P, Lee D (2023) Dynamic and stable hippocampal representations of social identity and reward expectation support associative social memory in male mice. *Nat Commun* **14**:2597
- Kouser M**, Speed HE, Dewey CM, Reimers JM, Widman AJ, Gupta N, Liu S, Jaramillo TC, Bangash M, Xiao B, *et al.* (2013) Loss of Predominant Shank3 Isoforms Results in Hippocampus-Dependent Impairments in Behavior and Synaptic Transmission. *The Journal of Neuroscience* **33**:18448-18468
- Lavi K**, Jacobson GA, Rosenblum K, Lüthi A (2018) Encoding of Conditioned Taste Aversion in Cortico-Amygdala Circuits. *Cell Rep* **24**:278-283
- Lee CCY**, Kheradpezhohu E, Diamond ME, Arabzadeh E (2020) State-Dependent Changes in Perception and Coding in the Mouse Somatosensory Cortex. *Cell Reports* **32**:108197
- Lee E**, Lee J, Kim E (2017) Excitation/Inhibition Imbalance in Animal Models of Autism Spectrum Disorders. *Biological Psychiatry* **81**:838-847
- Lee JY**, Jun H, Soma S, Nakazono T, Shiraiwa K, Dasgupta A, Nakagawa T, Xie JL, Chavez J, Romo R, *et al.* (2021) Dopamine facilitates associative memory encoding in the entorhinal cortex. *Nature* **598**:321-326
- Leekam SR**, Nieto C, Libby SJ, Wing L, Gould J (2007) Describing the Sensory Abnormalities of Children and Adults with Autism. *J Autism Dev Disord* **37**:894-910
- Levitan D**, Lin J-Y, Wachutka J, Mukherjee N, Nelson SB, Katz DB (2019) Single and population coding of taste in the gustatory cortex of awake mice. *J Neurophysiol* **122**:1342-1356
- Levitan D**, Liu C, Yang T, Shima Y, Lin J-Y, Wachutka J, Marrero Y, Ali Marandi Ghoddousi R, Veiga Beltrame E da, Richter TA, *et al.* (2020) Deletion of Stk11 and Fos in mouse BLA projection neurons alters intrinsic excitability and impairs formation of long-term aversive memory. *eLife* **9**:e61036 <https://doi.org/10.7554/eLife.61036>
- Li W-G**, Liu M-G, Deng S, Liu Y-M, Shang L, Ding J, Hsu T-T, Jiang Q, Li Y, Li F, *et al.* (2016) ASIC1a regulates insular long-term depression and is required for the extinction of conditioned taste aversion. *Nat Commun* **7**:13770

- Livneh Y**, Ramesh RN, Burgess CR, Levandowski KM, Madara JC, Fenselau H, Goldey GJ, Diaz VE, Jikomes N, Resch JM, *et al.* (2017) Homeostatic circuits selectively gate food cue responses in insular cortex. *Nature* **546**:611-616
- Marks TD**, Goard MJ (2021) Stimulus-dependent representational drift in primary visual cortex. *Nat Commun* **12**:5169
- Miura I**, Sato M, Overton ETN, Kunori N, Nakai J, Kawamata T, Nakai N, Takumi T (2020) Encoding of social exploration by neural ensembles in the insular cortex. *PLOS Biology* **18**:e3000584
- Monday HR**, Wang HC, Feldman DE (2023) Circuit-level theories for sensory dysfunction in autism: convergence across mouse models. *Front Neurol* **14**:1254297
- Monteiro P**, Feng G (2017) SHANK proteins: roles at the synapse and in autism spectrum disorder. *Nat Rev Neurosci* **18**:147-157
- Moran A**, Katz DB (2014) Sensory cortical population dynamics uniquely track behavior across learning and extinction. *J Neurosci* **34**:1248-1257
- Moreno-Bote R**, Beck J, Kanitscheider I, Pitkow X, Latham P, Pouget A (2014) Information-limiting correlations. *Nat Neurosci* **17**:1410-1417
- Ortiz-Cruz CA**, Marquez EJ, Linares-García CI, Perera-Murcia GR, Ramiro-Cortés Y (2022) Haploinsufficiency of Shank3 increases the orientation selectivity of V1 neurons. *Sci Rep* **12**:22230
- Pachitariu M**, Stringer C, Dipoppa M, Schröder S, Rossi LF, Dalglish H, Carandini M, Harris KD (2017) Suite2p: beyond 10,000 neurons with standard two-photon microscopy. :061507. Accessed February 25, 2022 <https://www.biorxiv.org/content/10.1101/061507v2>
- Pagano J**, Landi S, Stefanoni A, Nardi G, Albanesi M, Bauer HF, Pracucci E, Schön M, Ratto GM, Boeckers TM, *et al.* (2023) Shank3 deletion in PV neurons is associated with abnormal behaviors and neuronal functions that are rescued by increasing GABAergic signaling. *Molecular Autism* **14**:28
- Pal T**, Harvey S, Levine AS, Olszewski PK, Klockars A (2025) Neuromolecular Basis of Impaired Conditioned Taste Aversion Acquisition in Valproate-Induced Rat Model of Autism Spectrum Disorder. *Genes* **16**:203
- Pancholi R**, Ryan L, Peron S (2023) Learning in a sensory cortical microstimulation task is associated with elevated representational stability. *Nat Commun* **14**:3860
- Panzeri S**, Moroni M, Safaai H, Harvey CD (2022) The structures and functions of correlations in neural population codes. *Nat Rev Neurosci* **23**:551-567
- Peça J**, Feliciano C, Ting JT, Wang W, Wells MF, Venkatraman TN, Lascola CD, Fu Z, Feng G (2011) Shank3 mutant mice display autistic-like behaviours and striatal dysfunction. *Nature* **472**:437
- Poort J**, Khan AG, Pachitariu M, Nemri A, Orsolic I, Krupic J, Bauza M, Sahani M, Keller GB, Mrsic-Flogel TD, *et al.* (2015) Learning Enhances Sensory and Multiple Non-sensory Representations in Primary Visual Cortex. *Neuron* **86**:1478-1490
- Raymond MA**, Chapman IF, Staszko SM, Fletcher ML, Boughter JD (2024) Neural coding in gustatory cortex reflects consumption decisions: Evidence from conditioned taste aversion. :2024.02.28.582584. Accessed November 3, 2024 <https://www.biorxiv.org/content/10.1101/2024.02.28.582584v1>
- Raynaud F**, Janossy A, Dahl J, Bertaso F, Perroy J, Varrault A, Vidal M, Worley PF, Boeckers TM, Bockaert J, *et al.* (2013) Shank3-Rich2 interaction regulates AMPA receptor recycling and synaptic long-term potentiation. *J Neurosci* **33**:9699-9715
- Rodríguez-Durán LF**, López-Ibarra DL, Herrera-Xithe G, Bermúdez-Rattoni F, Osorio-Gómez D, Escobar ML (2023) Synergistic photoactivation of VTA-catecholaminergic and BLA-glutamatergic projections induces long-term potentiation in the insular cortex. *Neurobiology of Learning and Memory* **205**:107845
- Rodríguez-Durán LF**, Martínez-Moreno A, Escobar ML (2017) Bidirectional modulation of taste aversion extinction by insular cortex LTP and LTD. *Neurobiology of Learning and Memory* **142**:85-90

- Rosen AM**, Roussin AT, Di Lorenzo PM (2010) Water as an Independent Taste Modality. *Front Neurosci* **4**:175
- Rosenbaum R**, Smith MA, Kohn A, Rubin JE, Doiron B (2017) The spatial structure of correlated neuronal variability. *Nat Neurosci* **20**:107-114
- Rothschild G**, Nelken I, Mizrahi A (2010) Functional organization and population dynamics in the mouse primary auditory cortex. *Nat Neurosci* **13**:353-360
- Rumyantsev OI**, Lecoq JA, Hernandez O, Zhang Y, Savall J, Chrapkiewicz R, Li J, Zeng H, Ganguli S, Schnitzer MJ (2020) Fundamental bounds on the fidelity of sensory cortical coding. *Nature* **580**:100-105
- Sadacca BF**, Mukherjee N, Vladusich T, Li JX, Katz DB, Miller P (2016) The Behavioral Relevance of Cortical Neural Ensemble Responses Emerges Suddenly. *J Neurosci* **36**:655-669
- Samuelsen CL**, Gardner MPH, Fontanini A (2012) Effects of Cue-Triggered Expectation on Cortical Processing of Taste. *Neuron* **74**:410-422
- Sattin A**, Nardin C, Daste S, Moroni M, Reddy I, Liberale C, Panzeri S, Fleischmann A, Fellin T (2024) Aberration correction in long GRIN lens-based microendoscopes for extended field-of-view two-photon imaging in deep brain regions. Accessed January 9, 2025 <https://elifesciences.org/reviewed-preprints/101420>
- Scheerer NE**, Curcin K, Stojanoski B, Anagnostou E, Nicolson R, Kelley E, Georgiades S, Liu X, Stevenson RA (2021) Exploring sensory phenotypes in autism spectrum disorder. *Molecular Autism* **12**:67
- Schoonover CE**, Ohashi SN, Axel R, Fink AJP (2021) Representational drift in primary olfactory cortex. *Nature* **594**:541-546
- Sheintuch L**, Rubin A, Brande-Eilat N, Geva N, Sadeh N, Pinchasof O, Ziv Y (2017) Tracking the Same Neurons across Multiple Days in Ca²⁺ Imaging Data. *Cell Reports* **21**:1102-1115
- Staszko SM**, Boughter JD, Fletcher ML (2022) The impact of familiarity on cortical taste coding. *Curr Biol* **32**:4914-4924.
- Stringer C**, Pachitariu M, Steinmetz NA, Okun M, Bartho P, Harris KD, Sahani M, Lesica NA (2016) Inhibitory control of correlated intrinsic variability in cortical networks. *eLife* **5**:e19695 <https://doi.org/10.7554/eLife.19695>
- Tatavarty V**, Pacheco AT, Kuhnle CG, Lin H, Koundinya P, Miska NJ, Hengen KB, Wagner FF, Hooser SDV, Turrigiano GG (2020) Autism-Associated Shank3 Is Essential for Homeostatic Compensation in Rodent V1. *Neuron* **106**:769-777.
- Taylor JA**, Hasegawa M, Benoit CM, Freire JA, Theodore M, Ganea DA, Innocenti SM, Lu T, Gründemann J (2021) Single cell plasticity and population coding stability in auditory thalamus upon associative learning. *Nat Commun* **12**:2438
- Twarkowski H**, Steininger V, Kim MJ, Sahay A (2022) A dentate gyrus-CA3 inhibitory circuit promotes evolution of hippocampal-cortical ensembles during memory consolidation. *eLife* **11**:e70586 <https://doi.org/10.7554/eLife.70586>
- Vincis R**, Chen K, Czarnecki L, Chen J, Fontanini A (2020) Dynamic Representation of Taste-Related Decisions in the Gustatory Insular Cortex of Mice. *Current Biology* **30**:1834-1844.
- Vincis R**, Fontanini A (2016) Associative learning changes cross-modal representations in the gustatory cortex. *eLife* **5**
- Wang HC**, Feldman DE (2024) Degraded tactile coding in the Cntnap2 mouse model of autism. *Cell Reports* **43**:114612
- Wang L**, Gillis-Smith S, Peng Y, Zhang J, Chen X, Salzman CD, Ryba NJP, Zuker CS (2018) The coding of valence and identity in the mammalian taste system. *Nature* **558**:127-131
- Wang PY**, Boboila C, Chin M, Higashi-Howard A, Shamash P, Wu Z, Stein NP, Abbott LF, Axel R (2020) Transient and Persistent Representations of Odor Value in Prefrontal Cortex. *Neuron* **108**:209-224.

- Wang X**, McCoy PA, Rodriguiz RM, Pan Y, Je HS, Roberts AC, Kim CJ, Berrios J, Colvin JS, Bousquet-Moore D, *et al.* (2011) Synaptic dysfunction and abnormal behaviors in mice lacking major isoforms of Shank3. *Human Molecular Genetics* **20**:3093-3108
- Wen W**, Turrigiano GG (2021) Developmental Regulation of Homeostatic Plasticity in Mouse Primary Visual Cortex. *J Neurosci* **41**:9891-9905
- Woods NI**, Stefanini F, Apodaca-Montano DL, Tan IMC, Biane JS, Kheirbek MA (2020) The Dentate Gyrus Classifies Cortical Representations of Learned Stimuli. *Neuron* **107**:173-184.
- Wu C-H**, Ramos R, Katz DB, Turrigiano GG (2021) Homeostatic synaptic scaling establishes the specificity of an associative memory. *Curr Biol* **31**:2274-2285.
- Yiannakas A**, Kolatt Chandran S, Kayyal H, Gould N, Khamaisy M, Rosenblum K (2021) Parvalbumin interneuron inhibition onto anterior insula neurons projecting to the basolateral amygdala drives aversive taste memory retrieval. *Curr Biol* **31**:2770-2784.
- Yiannakas A**, Rosenblum K (2017) The Insula and Taste Learning. *Front Mol Neurosci*. Accessed January 14, 2021 <https://www.frontiersin.org/articles/10.3389/fnmol.2017.00335/full>
- Zhang X**, Li B (2018) Population coding of valence in the basolateral amygdala. *Nat Commun* **9**:5195
- Zhang Y**, *et al.* (2023) Fast and sensitive GCaMP calcium indicators for imaging neural populations. *Nature* **615**:884-891
- Zhao M**, Kwon SE (2023) Interneuron-Targeted Disruption of *SYNGAP1* Alters Sensory Representations in the Neocortex and Impairs Sensory Learning. *J Neurosci* **43**:6212-6226

Peer reviews

Reviewer #1 (Public review):

Summary:

The study from Wu and Turrigiano investigates how disruption of taste coding in a mouse model of autism spectrum disorders (ASDs) affects aversive learning in the context of a conditioned taste aversion (CTA) paradigm. The experiments combine 2-photon calcium imaging of neurons in the gustatory portion of the anterior insular cortex (i.e., gustatory cortex) with behavioral training and testing. The authors rely on Shank3 knockout mice as a model for ASDs. The authors found that Shank3 mice learn CTA more slowly and extinguish the memory more rapidly than control subjects. Calcium imaging identified impairments in taste-evoked activity associated with memory encoding and extinction. During memory encoding, the authors found less suppressed neuronal activity and increased correlated variability in Shank3 mice compared to controls. During extinction, they observed a faster loss of taste selectivity and degradation of taste discriminability in mutants compared to controls.

Strengths:

This is a well-written manuscript that presents interesting findings. The results on the learning and extinction deficits in Shank3 mice are of particular interest. Analyses of neural activity are well conducted and provide important information on the type of impaired cortical activity that may correlate with behavioral deficits.

Weaknesses:

- (1) The experiments rely on three groups: CS-only WT, CTA WT, and CTA KO. Can the authors provide a rationale for not having a CS-only KO group?
- (2) The authors design an effective behavioral paradigm comparing consumption of water and saccharin and tracking extinction (Figure 3). This paradigm shows differences in licking

across distinct behavioral conditions. For instance, during T1, licking to water strongly differs from licking to saccharin for both WT and KO. During T2, licking to water strongly differs from licking to saccharin only for WT (much less for KO), and licking to saccharin in WT differs from that in KO. These differences in taste sampling across conditions could contribute to some of the effects on neural activity and discriminability reported in Figures 5 and 6. That is sucrose and water trials may be highly discriminable because in one case the mouse licks and in the other it does not (or licks much less). The author may want to address this issue.

(3) Are there any omission trials following CTA? If so, they should be quantified and reported. How are the omission trials treated with regard to the analyses?

(4) The authors describe the extinction paradigm as "alternative choice". In decision-making, alternative choice paradigms typically require 2 lateral spouts to report decisions following the sampling from a central spout. To avoid confusion, the authors may want to define their paradigm as alternative sampling.

(5) Figure 4 reports that CTA increases the proportion of neurons that consistently respond to saccharin and water across days. While the saccharin result could be an effect of aversive learning, it is less clear why the phenomenon would generalize to water as well. Can the authors provide an explanation?

(6) The recordings are performed in the part of the anterior insular cortex that is typically defined as "gustatory cortex" (GC). Given the functional heterogeneity of the anterior insular cortex (AIC) and given that the authors do not sample all of the anteroposterior extent of AIC, I would suggest being more explicit about their positioning in GC. Also, some citations (e.g., Gogolla et al, 2014) refer to the posterior insular cortex, which is considered more inherently multimodal than GC. GC multimodality is typically associative in nature, as only a few neurons respond to sound and light in naïve animals.

(7) It would be useful to add summary figures showing the extent of viral spread as well as GRIN lens placement.

(8) I encourage the authors to add Ns every time percentages are reported. How many neurons have been recorded in each condition? Can the authors provide the average number of neurons recorded per session and per animal?

(9) It looks like some animals learned more than others (Figure 1E or Figure 3C). Is it possible to compare neural activity across animals that showed different degrees of learning?

<https://doi.org/10.7554/eLife.108988.1.sa2>

Reviewer #2 (Public review):

Wu and Turrigiano investigated how cortical taste coding during conditioned taste aversion (CTA) learning is affected in Shank3 knockout (KO) mice, a model of monogenic ASD. Using longitudinal two-photon calcium imaging of AIC neurons, the authors show that Shank3 KO mice exhibit reduced suppression of activity in a subset of neurons and a higher correlated variability in neural activity. This is accompanied by slower learning and faster extinction of aversive taste memories. These results suggest that Shank3 loss compromises the flexibility and stability of cortical representations underlying adaptive behaviour.

Major strengths:

(1) Conceptual significance: The study connects a molecular ASD risk gene (Shank3) to flexible sensory encoding, bridging genetics, systems neuroscience, and behaviour.

(2) Technical rigour: Longitudinal calcium imaging with cell-registration across learning and extinction sessions is technically demanding and well-executed.

(3) Behavioural paradigm: The use of both acquisition and extinction paradigms provides a more nuanced picture of learning dynamics.

(4) Analyses: Correlated variability, discriminability indices, and population decoding analyses are robust and appropriate for addressing behavioural and network-level coding changes.

Major weaknesses:

(1) Causality: The paper infers that increased correlated variability causes learning deficits, but no causal tests (e.g., optogenetic modulation of inhibition or interneuron rescue) are presented to confirm this.

(2) Behavioural scope: The study focuses exclusively on taste aversion; generalisation to other flexible learning paradigms (e.g., reversal or probabilistic tasks) is not addressed.

(3) Mechanistic insights: While providing interesting findings of altered sensory perception and extinction of learning-related signals in AIC, it offered nearly no mechanistic insights. This makes the interpretation, especially on how generalisable these findings are, difficult. Also, different reported findings are "potentially" connected, but the exact relation between increased correlated variability and faster loss of taste selectivity cannot be assessed.

<https://doi.org/10.7554/eLife.108988.1.sa1>

Reviewer #3 (Public review):

In this study, Wu & Turrigiano investigate an ethologically relevant form of associative learning (conditioned taste aversion - CTA) and its extinction in the Shank3 KO mouse model of ASD. They also examine the underlying circuits in the anterior insular cortex (AIC) simultaneously, using two-photon calcium imaging through a GRIN lens. They report that Shank3 KO mice learn CTA slower and suggest that this is mediated by a reduction in tastant-stimulus activity suppression of AIC neurons and a reduced signal-to-noise ratio due to increased noise correlations in AIC neurons. Interestingly, once Shank3 KO mice acquire CTA, they extinguish the aversive memory more rapidly than wild-type mice. This accelerated extinction is accompanied by a faster loss of neuronal and population-level taste selectivity and coding in the AIC compared to WT mice.

This is an important study that uses in vivo methods to assess circuit dysfunction in a mouse model of ASD, related to sensory perception valence (in this case, taste). The study is well executed, the data are of high quality, and the analytical procedures are detailed. Furthermore, the behavioural paradigm is well thought out, particularly the approach for assessing extinction through repeated retrieval sessions (T1-T5), which effectively tests discrimination between saccharin and water rather than relying solely on lick counts or total consumption as a measure of extinction. Finally, the statistical tests used are appropriate and justified.

There is, however, a missing link between the behavioural findings and the underlying mechanisms. More specifically:

(1) The authors don't make a causal link between the behaviour and AIC neurophysiology, both the percentage of suppressed cells and the coactivity measurements. For the % of suppressed cells, it seems that both WT and KO cells are suppressed in the transition between CST1 and CST2 (Figure 1L), yet only the WT mice exhibit CTA (at least by CST2). For the taste-

elicited coactivity measure, it seems that there is an increase in coactivity from CST1 to CST2 in WT (Figure 2C - blue, although not statistically tested?), but persistently higher coactivity in KO. Is this change of coactivity in WT important for the expression of CTA? Plotting behavioral performance (from Figure 1G) against coactivity (from Figure 2C) for each animal would be informative.

(2) Shank3 KO cells already show an increase in baseline coactivity (Figure 2- figure supplement 1), and the authors never examine CS-only responses in the KO group, therefore making it difficult to determine whether elevated coactivity and noise correlations reflect a generalized AIC abnormality in Shank3 KOs (perhaps through impaired PV-mediated inhibition in insular cortex - Gogolla et al, 2014) that is not directly responsible/related to CTA?

(3) How do the authors interpret the large range of lick ratios (Figure 1G) for WT (almost bi-modal distribution)? Is there a within-subject correlation with any of the neurophysiological measurements to suggest a relationship between AIC neurophysiology and behavioural expression of CTA?

(4) Indeed, CTA appears to be successfully achieved for Shank3 KO mice delayed by 1 day, as the level of saccharin aversion during the first retrieval session (T1) is comparable between Shank3 KO and WTs. In this context, not extending the first part of the paradigm to include CST3 seems to be a missed opportunity. Doing so would have allowed for within-cell and within-subject comparison of taste-elicited pairwise correlation across the learning and to investigate the neural mechanism of delayed extinction in KOs more effectively.

(5) How to interpret Figure 5F: Absolute discriminability is lower for T5 for CTA WT and CTA KO compared to CS-only? Why would AIC neurons have less information on taste identity by the end of extinction than during the unconditioned (CS-only) condition? And if that is the case, how is decoding accuracy in Figure 6C higher in T5 for CTA WT vs CS-only?

<https://doi.org/10.7554/eLife.108988.1.sa0>